

ON RINGS OF OPERATORS. II*

BY

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Introduction. This paper is a continuation of one by the same authors: *On rings of operators*, Annals of Mathematics, (2), vol. 37 (1936), pp. 116–229. It contains the solution of certain problems which, were left open there. We will prove the general additivity of trace $Tr_M(A)$, its weak continuity, and certain isomorphisms between $\mathfrak{H}$, M , and M' (cf. the remarks (i)–(iv) at the end of the above quoted paper). All these considerations refer to “Case (II)” for M (cf. Theorem VIII, loc. cit.).

The properties of $Tr_M(A)$ are established by obtaining for it a representation

$$Tr_M(A) = \sum_{i=1}^m (Ag_i, g_i)$$

(with a fixed, finite $m = 1, 2, \dots$, and fixed $g_1, \dots, g_m \in \mathfrak{H}$). This representation is remarkable, because it is obviously a close analogue of the representation of $Tr_M(A)$ as a trace, that is, as the arithmetic mean of the diagonal matrix-elements of A in the cases (I_n) , $n = 1, 2, \dots$, when M is essentially the full matrix ring of an n -(finite-) dimensional Euclidean space.

For certain cases (with the help of which the others are then mastered) we have even $m = 1$.

In [Part I](#) the above representation of $Tr_M(A)$ is obtained approximately. The technically interested reader may find it worth observing that the exhaustion method we use there ([§§1.2 and 1.3](#)) is analogous to certain procedures which can be used advantageously in the theories of measures and integration too. On this basis we establish the main properties of $Tr_M(A)$ in [Part II](#), and then obtain the exact representation of $Tr_M(A)$ in [Part III](#). Here two maximum-problems, called (A) and (B), which seem to possess some independent interest too, play a decisive role.

[Part IV](#) is devoted to establishing an isomorphism between $\mathfrak{H}$, M , and M' . It turns out that a certain algebraic-topological extension $Q(M)$ of M is isomorphic to $\mathfrak{H}$ and that M and M' play in it the role of right- and left-multiplication. This leads to an interesting and entirely new type of infinite hypercomplex systems, which are at the same time Hilbert spaces. A subsequent paper will be devoted to their independent study.

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The appendix deals with the possibility of considering M (in the case (II_1)) as a system of matrices with continuously spread rows and columns.

We will use the notations, definitions, and results of our paper *On rings of operators*, quoted above, throughout this paper. We will quote it, whenever necessary, as **R.O.** All other quotations follow the bibliography of **R.O.** (pp. 125–126, Nos. (1)–(22)).

The isomorphism problems of different rings M of class (II_1) (cf. the remark (v) at the end of **R.O.**) are not discussed here. They will be dealt with in a subsequent publication.

Since the appearance of **R.O.** the second-named author has succeeded in finding new representations of case (II_1) in terms of infinite direct products, which throw new light on (II_1) as a limiting case of the (I_n) , $n = 1, 2, \dots$, as well as a way of applying the present theory to quantum-mechanics. These subjects too will be discussed in papers which will follow soon.

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CHAPTER I. APPROXIMATE FORM OF $Tr_M(A)$ (FOR $\alpha \geq 1$)

1.1. We assume that we have given a factor M in case II_1 . Now we normalize D_M and $D_{M'}$ so that $C = 1$ (cf. [R.O.](#), pp. 179–182). This means ([R.O.](#), loc. cit.) that for every $f \in \mathfrak{H}$, $D_M(\mathfrak{M}_f^{M'}) = D_{M'}(\mathfrak{M}_f^M)$ and that the range of D_M is the interval $0 \leq x \leq 1$. Let the range of $D_{M'}$ be the interval $0 \leq x \leq \alpha$, where $0 < \alpha \leq \infty$. In this part, we assume that $\alpha \geq 1$.

Now M' is a factor of class II_1 or II_∞ , by [R.O.](#), Theorem X. In the first case the range of $D_{M'}$ is in the normalization (loc. cit.) the interval, $0 \leq x \leq 1$, and $C = 1/\alpha$ (since $D_{M'} = 1/\alpha$ times its value in the previous paragraph). Since $C \leq 1$, we have in both cases by the discussion on page 182 of [R.O.](#), that $\Delta_0 = \Delta$. By [R.O.](#), Definition 10.1.1, this implies that there exists an f such that $D_M(\mathfrak{M}_f^{M'}) = 1$. Since we can pass from the normalization of this paragraph to that of the previous paragraph by multiplying $D_{M'}$ by α and leaving D_M unchanged, we see that this holds even with our previous normalization.

Thus we have an f such that $D_M(E_f^{M'}) = 1 = D_M(1)$ or $D_M(1 - E_f^{M'}) = 0$ which implies $1 - E_f^{M'} = 0$; that is, $E_f^{M'} = 1$. We now assume that such an f has been chosen and is held fixed for the following discussion.

1.2. With respect to this fixed f , we now define certain relations concerning a projection $E \in M$.

Definition 1.2.1. Let $E \neq 0$ be a projection, $\theta \in M$, θ a real number ≥ 0 . The relation $E > \theta$ ($E \geq \theta$, $E < \theta$, $E \leq \theta$) is said to hold if $(Ef, f) > \theta D_M(E)$ ($(Ef, f) \geq \theta D_M(E)$, $(Ef, f) < \theta D_M(E)$, $(Ef, f) \leq \theta D_M(E)$) respectively.

The relation $E \underset{p}{\geq} \lambda$ ($E \underset{p}{\leq} \lambda$) is said to hold for a projection $E \neq 0$ if for every $F \in M$

such that $F \leq E$, $F \neq 0$, we have $F \geq \lambda$ ($F \leq \lambda$).

Lemma 1.2.1. Let $\{E_i\}$ be a sequence of projections (finite or infinite) with $E_i \in M$, $E_i E_j = \delta_{ij}^i E_i$. Then if for every i , $E_i \leq \lambda$ ($E_i \geq \lambda$), and if for some i , say i_0 , $E_{i_0} < \lambda$ ($E_{i_0} > \lambda$) we have $\sum_i E_i < \lambda$ ($\sum_i E_i > \lambda$).

We have, for every i , $\lambda D(E_i) \leq (E_i f, f)$, hence $\lambda \sum_{i \neq i_0} D(E_i) \leq \sum_{i \neq i_0} (E_i f, f)$ and $\lambda D(E_{i_0}) < (E_{i_0} f, f)$. These imply $\lambda \sum_i D(E_i) < \sum_i (E_i f, f)$ or $\lambda D(\sum_i E_i) < (\sum_i E_i f, f)$, which was to be shown.

Lemma 1.2.2. If $E \geq \lambda$ ($E \leq \lambda$), then there exists an F such that $E \geq F$, $F \geq \lambda$ ($F \leq \lambda$).

By [Definition 1.2.1](#) we have $E \neq 0$. Suppose there is no such F . Then there must be an E_1 , such that $E \geq E_1$, and $E_1 < \lambda$. (Otherwise E itself would be such an F .) Now let Ω be the first ordinal number which belongs to a cardinal number $\aleph > \aleph_0$. Let α be an ordinal number such that $\alpha < \Omega$. Let us suppose that for all $\beta < \alpha$ an E_β has been defined in such a way that $E_\beta \neq 0$, $E_\beta \leq E$, $E_\beta < \lambda$, $E_{\beta_1} E_{\beta_2} = \delta_{\beta_1, \beta_2} E_{\beta_1}$. Now if $E - \sum_{\beta < \alpha} E_\beta = 0$, let $E_{\alpha'}$ be undefined for $\alpha' \geq \alpha$. If, however, $E - \sum_{\beta < \alpha} E_\beta \neq 0$, by hypothesis, $E - \sum_{\beta < \alpha} E_\beta$ is not $\geq \lambda$, hence there exists an E_0 , with $E_0 \neq 0$, $E - \sum_{\beta < \alpha} E_\beta \geq E_0$, $E_0 < \lambda$, $E \geq E - \sum_{\beta < \alpha} E_\beta \geq E_0$. Thus if we let $E_\alpha = E_0$, we have, if $E - \sum_{\beta < \alpha} E_\beta \neq 0$, an E_α which is orthogonal to all previous E_β and $E_\alpha < \lambda$.

Now $D(E) \geq \sum_{\beta < \alpha} D(E_\beta)$ for every α . Since $D(E_\beta) > 0$, this implies that there is only denumerably many of the numbers $D(E_\beta)$. Hence for some $\alpha < \Omega$, $E - \sum_{\beta < \alpha} E_\beta = 0$. Inasmuch as $\alpha < \Omega$, we can re-index the E_β 's into a finite or a simply infinite sequence. Now $E_1 < \lambda$, $E_i \leq \lambda$, and since $E = \sum_i E_i$, [Lemma 1.1](#) implies that $E < \lambda$, a contradiction.

Lemma 1.2.3. If $E \geq \lambda$ ($E \leq \lambda$), then if $E \geq F$, $F \neq 0$, $F \in M$, we have $F \geq \lambda$ ($F \leq \lambda$).

If F_1 is such that $F \geq F_1$, $F_1 \neq 0$ then $E \geq F \geq F_1$ and hence by [Definition 1.2.1](#), $F_1 \geq \lambda$. This statement implies that $F \geq \lambda$.

1.3. Now $(1f, f) = \|f\|^2 = \|f\|^2 D(1)$. Now let $\theta_0 = \|f\|^2$ which is of course not zero for $\mathfrak{M}_f^M = \mathfrak{H}$ and this precludes $f = 0$. Thus 1 (the projection, not the number) is $\geq \theta_0$. By [Lemma 1.2.2](#), there exists an $E \in M$, such that $E \geq \theta_0$. Now let λ be the least upper bound of the numbers θ such that $E \geq \theta$. λ must be less than $(Ef, f)/D_M(E)$ and since $D_M(E) > 0$, it must be finite. Let ϵ be any number > 0 . E is not $\geq \lambda + \epsilon$. Hence there is an E_1 , such that $E \geq E_1$, and $E_1 \leq \lambda + \epsilon$.

Lemma 1.2.2 now implies that there is an E_2 with $E_1 \geq E_2$ and $E_2 \leq \lambda + \epsilon$.

Now if F is such that $E \geq F$, $F \in M$, $F \neq 0$, we have $F \geq \theta$ for all $\theta < \lambda$, which means of course that $F \geq \lambda$ and $E \geq \lambda$. Now since $E \geq E_1 \geq E_2$, **Lemma 1.2.3** implies that $E_2 \geq \lambda$. Thus for some fixed $\lambda > 0$, given any $\epsilon > 0$, we can find a non-zero $E_2 \in M$, such that for all $F \in M$ with the property $E_2 \geq F$, we have

$$(\lambda + \epsilon)D_M(F) \geq (Ff, f) \geq \lambda D_M(F).$$

Now if we let f be $\lambda^{1/2}f$ above, $\epsilon = \lambda(K - 1)$, $E_2 = E$, we see that

Lemma 1.3.1. *To every $K > 1$ there exists an f and $E \in M$, $E \neq 0$, such that for every $F \in M$ with the property $E \geq F$,*

$$KD_M(F) \geq (Ff, f) \geq D_M(F).$$

1.4. We can now show

Lemma 1.4.1. *Let K , f , and E be as in **Lemma 1.3.1**. Let $A \in M$ be a positive definite self-adjoint operator such that $EA = AE = A$. Then*

$$KTr_M(A) \geq (Af, f) \geq Tr_M(A).$$

Let c be the bound of A , $E(\lambda)$ the resolution of the identity corresponding to A . Now by **R.O.**, pp. 212–213,

$$Tr_M(A) = \int_0^c \lambda dD(E(\lambda)); \quad (Af, f) = \int_0^c \lambda d(E(\lambda)f, f).$$

Now since $AE = A$, $E(0) \geq 1 - E$, and since A commutes with E , E commutes with all $E(\lambda)$ (cf. (15), Theorem 8.2). Now $E(\lambda) = E(\lambda)E + E(\lambda)(1 - E)$. Since $E(\lambda)$ and E commute, $E(\lambda)E = F(\lambda)$ is a projection and similarly $E(\lambda)(1 - E)$ is a projection. For $\lambda \geq 0$, we have $E(\lambda)(1 - E) \leq 1 - E$ and also $E(\lambda)(1 - E) \geq E(0)(1 - E) \geq (1 - E)^2 = 1 - E$ and thus $E(\lambda)(1 - E) = 1 - E$. So for $\lambda \geq 0$, $E(\lambda) = F(\lambda) + (1 - E)$. Now $F(\lambda)(1 - E) = E(\lambda)E(1 - E) = 0$, hence by **R.O.**, Definition 8.2.1, $D(E(\lambda)) = D(F(\lambda)) + D(1 - E)$. Thus we have

$$\begin{aligned} Tr_M(A) &= \int_0^c \lambda dD(E(\lambda)) = \int_0^c \lambda dD(F(\lambda)) + \int_0^c \lambda dD(1 - E) \\ &= \int_0^c \lambda dD(F(\lambda)) \end{aligned}$$

and

$$\begin{aligned}
 (Af, f) &= \int_0^c \lambda d(E(\lambda)f, f) \\
 &= \int_0^c \lambda d(F(\lambda)f, f) + \int_0^c \lambda d((1-E)f, f) \\
 &= \int_0^c \lambda d(F(\lambda)f, f).
 \end{aligned}$$

But [Lemma 1.3.1](#) now implies that if $0 \leq \alpha < \beta \leq c$, then

$$KD_M(F(\beta) - F(\alpha)) \geq ((F(\beta) - F(\alpha))f, f) \geq D_M(F(\beta) - F(\alpha))$$

which by the definition of the Riemann-Stieltjes integral yields that

$$KTr_M(A) \geq (Af, f) \geq Tr_M(A).$$

Lemma 1.4.2. *For a projection $E \neq 0$, $E \geq \theta$ ($E \leq \theta$) is equivalent to the statement that for all positive definite $A \in M$ such that $\overset{p}{EA} = \overset{p}{AE} = A$,*

$$(Af, f) \geq \theta Tr_M(A) \quad (\leq \theta Tr_M(A)).$$

Since $Tr_M(F) = D_M(F)$, F a projection, the last statement implies the first. The converse is shown by a proof similar to that of [Lemma 1.4.1](#).

Lemma 1.4.3. *Let $A \in M$ be such that $EA = AE = A$, then $\|A^*f\|^2 \leq K\|Af\|^2$, if E , f , K are as in [Lemma 1.3.1](#).*

A^*A is positive definite and $\in M$. Furthermore $AE = EA = A$ implies $EA^* = A^*E = A^*$, and thus $EA^*A = A^*A = A^*AE$. Hence [Lemma 1.4.1](#) applies to A^*A and we have

$$(\alpha) \quad KTr_M(A^*A) \geq (A^*Af, f) \geq Tr_M(A^*A).$$

Using the canonical decomposition ([R.O.](#), Definition 4.4.1) for A^* we have $A^* = UB$, where U may be taken as unitary. (In the finite cases, we see (cf. [R.O.](#), Lemma 16.1.1) that there exists a partially isometric V with initial set $(f; A^*f = 0)$ and final set $(f; Af = 0)$. Now if W is as in [R.O.](#), Definition 4.4.1, for $A = A^*$, let $U = W + V$.) B is self-adjoint and equals $(AA^*)^{1/2}$. Now $A = BU^* = BU^{-1}$ and hence $A^*A = UBBU^{-1} = UB^2U^{-1} = UAA^*U^{-1}$. Hence

$$(\beta) \quad Tr_M(A^*A) = Tr_M(UAA^*U^{-1}) = Tr_M(AA^*).$$

Substituting A^* for A in our previous result we have

$$(\gamma) \quad KTr_M(AA^*) \geq (AA^*f, f) \geq Tr_M(AA^*).$$

Combining (α) , (β) , and (ϕ) , we obtain

$$K(A^*Af, f) \geq KTr_M(A^*A) = KTr_M(AA^*) \geq (AA^*f, f).$$

Since $(A^*Af, f) = (Af, Af) = \|Af\|^2$, $(AA^*f, f) = (A^*f, A^*f) = \|A^*f\|^2$; this is the desired inequality.

1.5. Now if E is as in [Lemma 1.3.1](#), let n be the smallest integer such that $1/n \leq D_M(E)$. Then if $E^0 \leq E$ is such that $D_M(E^0) = 1/n$, [Lemma 1.3.1](#) will hold with E^0 in place of E . Now f was chosen in such a manner that $\mathfrak{M}_f^{M'} = \mathfrak{H}$. This implies that $\mathfrak{M}_{E^0f}^{M'}$ is the range of E^0 , because since the set $(A'f; A' \in M')$ is dense in $\mathfrak{H}$, the set $(E^0A'f; A' \in M') = (A'E^0f; A' \in M')$ must be dense in the range of E^0 , or $E^0 = E_{E^0f}^{M'}$. Since $C = 1$, $D_{M'}(E_{E^0f}^{M'}) = D_M(E_{E^0f}^{M'}) = D_M(E^0) = 1/n$.

Now let $E^0 = E_1$, $E_{E^0f}^{M'} = E'_1$. Since $D_M(E_1) = 1/n$, $D_M(1) = 1$, we can find n projections $\{E_j\}$, $j = 1, \dots, n$ with the first equal to E_1 such that $E_j \in M$, $\sum_{j=1}^n E_j = 1$, $E_i E_j = 0$ if $i \neq j$, $D_M(E_j) = 1/n$. Since $D_{M'}(E'_1) = 1/n$, $D_{M'}(1) = \alpha \geq 1$, there exist n projections $E'_j \in M'$, $j = 1, \dots, n$, with the first equal to E'_1 and such that $E'_i E'_j = 0$ if $i \neq j$ and $D_{M'}(E'_j) = 1/n$. Since $D_M(E_1) = D_M(E_j)$, there is a partially isometric operator $W_j \in M$ with initial set the range of E_1 and final set the range of E_j (cf. [R.O.](#), Definition 4.3.1). Similarly we have a $W'_j \in M'$ with initial set the range of E'_1 and final set E'_j . The relations $W_j^* W_j = E_1$, $W_j W_j^* = E_j$, $W_j'^* W'_j = E'_1$ and $W'_j W_j'^* = E'_j$ hold, also $W_j E_1 = W_j$, $E_j W_j = W_j$, $E_1 W_j^* = W_j^*$, $W_j^* E_j = W_j^*$, $W_j' E'_1 = W_j'$, $E'_j W'_j = W_j'$, $E'_1 W_j'^* = W_j'^*$, $W_j'^* E'_j = W_j'^*$ (cf. [R.O.](#), Lemma 4.3.1).

Let $W_j' W_j E_1 f = f_j$, $E_1 f = f_1$. Then $E_j f_j = E_j W_j' W_j E_1 f = W_j' E_j W_j E_1 f = W_j' W_j E_1 f = f_j$ and $E'_j f_j = E'_j W_j' W_j E_1 f = W_j' W_j E_1 f = f_j$. Thus $E_j f_j = f_j = E'_j f_j$.

Let $g = \sum_{j=1}^n f_j$, then $E_i g = E_i \sum_{j=1}^n f_j = E_i \sum_{j=1}^n E_j f_j = \sum_{j=1}^n E_i E_j f_j = E_i f_i = f_i$. Similarly $E'_i g = f_i$. Now for any $A \in M$ if either $i \neq j$ or $k \neq l$, then $E_i A E_k g$ is orthogonal to $E_j A E_l g$. For inasmuch as $E_k g = f_k = E'_k g$, we have,

$$\begin{aligned} (E_i A E_k g, E_j A E_l g) &= (E_j E_i A E_k g, A E_l g) = \delta_j^i (E_i A E_k g, A E_l g) \\ &= \delta_j^i (E_i A E'_k g, A E'_l g) = \delta_j^i (E'_k E_i A g, E'_l A g) \\ &= \delta_j^i (E'_l E'_k E_i A g, A g) = \delta_j^i \delta_l^k (E'_k E_i A g, A g). \end{aligned}$$

These results imply that if $A \in M$, then

$$\|Ag\|^2 = \left\| \left(\sum_{i=1}^n E_i \right) A \left(\sum_{j=1}^n E_j \right) g \right\|^2$$

$$\begin{aligned}
&= \left\| \sum_{i=1}^n \sum_{j=1}^n E_i A E_j g \right\|^2 = \sum_{i=1}^n \sum_{j=1}^n \|E_i A E_j g\|^2 \\
&= \sum_{i=1}^n \sum_{j=1}^n \|E_i A E_j E_j g\|^2 = \sum_{i=1}^n \sum_{j=1}^n \|E_i A E_j f_j\|^2 \\
&= \sum_{i=1}^n \sum_{j=1}^n \|E_i A E_j W_j' W_j f_1\|^2 \\
&= \sum_{i=1}^n \sum_{j=1}^n \|W_j' E_i A E_j W_j f_1\|^2 \\
&= \sum_{i=1}^n \sum_{j=1}^n \|W_j' E_1' E_i A E_j W_j f_1\|^2 \\
&= \sum_{i=1}^n \sum_{j=1}^n \|E_1' E_i A E_j W_j f_1\|^2 \\
&= \sum_{i=1}^n \sum_{j=1}^n \|E_i A E_j W_j E_1' f_1\|^2 \\
&= \sum_{i=1}^n \sum_{j=1}^n \|E_i A E_j W_j f_1\|^2 \\
&= \sum_{i=1}^n \sum_{j=1}^n \|W_i^* E_i A E_j W_j f_1\|^2 \\
&= \sum_{i=1}^n \sum_{j=1}^n \|W_i^* E_i A E_j W_j E_1 f\|^2 \\
&= \sum_{i=1}^n \sum_{j=1}^n \|W_i^* E_i A E_j W_j f\|^2,
\end{aligned}$$

remembering that W_j' is isometric on the range of E_1' , W_i^* on the range of E_i and $W_j E_1 = W_j$. Substituting A^* for A and interchanging i and j we also have

$$\|A^* g\|^2 = \sum_{i=1}^n \sum_{j=1}^n \|W_j^* E_j A^* E_i W_i f\|^2$$

$$= \sum_{i=1}^n \sum_{j=1}^n \|(W_i^* E_i A E_j W_j)^* f\|^2.$$

But recalling the properties of W_j and W_i^* we have

$$E_1 W_i^* E_i A E_j W_j = W_i^* E_i A E_j W_j = W_i^* E_i A E_j W_j E_1.$$

Thus [Lemma 1.4.3](#) with $A = W_i^* E_i A E_j W_j$ yields $\|(W_i^* E_i A E_j W_j)^* f\|^2 \leq K \|W_i^* E_i A E_j W_j f\|^2$ and with this, we obtain from the above equations for $\|Ag\|^2$ and $\|A^*g\|^2$ that $\|A^*g\|^2 \leq K \|Ag\|^2$.

This result may be stated as follows.

Lemma 1.5.1. *If K is any number > 1 , there exists a $g \in \mathfrak{H}$, $g \neq 0$ such that for every $A \in M$, $\|A^*g\|^2 \leq K \|Ag\|^2$. Obviously we may assume that $\|g\| = 1$.*

1.6. We now have

Lemma 1.6.1. *Let K and g be as in [Lemma 1.5.1](#). Then if E and $F \in M$, are two projections such that $D_M(E) = D_M(F)$, then $K^{-1}(Fg, g) \leq (Eg, g) \leq K(Fg, g)$.*

Inasmuch as $D_M(E) = D_M(F)$, there exists a partially isometric operator W such that $W^*W = F$, $WW^* = E$, $W \in M$ (cf. [R.O.](#), Definition 8.2.1, Definition 6.1.1, and Lemma 4.3.1). Now $(Fg, g) = (W^*Wg, g) = (Wg, Wg) = \|Wg\|^2$, $(Eg, g) = (WW^*g, g) = (W^*g, W^*g) = \|W^*g\|^2$. [Lemma 1.5.1](#) with $A = W$ implies $(Eg, g) \leq K(Fg, g)$. With $A = W^*$, the same lemma implies $(Fg, g) \leq K(Eg, g)$ or $K^{-1}(Fg, g) \leq (Eg, g)$. We have now shown our lemma.

Suppose $E \in M$ is such that $D_M(E) = 1/m$, where m is an integer. Then there exist $m-1$ projections $E_2, \dots, E_m$, such that when we let $E_1 = E$, we have $\sum_{j=1}^m E_j = 1$, $E_i E_j = 0$, for $i \neq j$, $E_j \in M$, $D_M(E_j) = 1/m$. Now returning to the g of [Lemmas 1.5.1](#) and [1.6.1](#), we have

$$1 = \|g\|^2 = (g, g) = \left(\sum_{j=1}^m E_j g, g \right) = \sum_{j=1}^m (E_j g, g).$$

Let $(E_1 g, g) = \alpha$. Then [Lemma 1.6.1](#) implies that $K\alpha \geq (E_j g, g) \geq K^{-1}\alpha$, for every j . Summing over j , gives $Km\alpha \geq 1 \geq K^{-1}m\alpha$ or $K\alpha \geq 1/m \geq K^{-1}\alpha$ which is the same as

$$(+) \quad K(Eg, g) \geq D_M(E) \geq K^{-1}(Eg, g),$$

since $D_M(E) = 1/m$.

Let us study the class Σ of E 's in M which satisfy the equation (+). Now if $F_1, \dots, F_q$ or $F_1, F_2, \dots$ satisfy (+) and $F_i F_j = 0$, $i \neq j$, then $\sum_{j=1}^q F_j$ or $\sum_{j=1}^{\infty} F_j$

also satisfies (+). But if $F \in M$ is such that $D_M(F) = q/p$, $F = \sum_{j=1}^q F_j$ for mutually orthogonal F_j 's $\in M$ with $D_M(F_j) = 1/p$ and hence satisfying (+). Thus if $D_M(F) = q/p$, F satisfies (+).

Let E be any projection of M , $D_M(E) = \alpha$. Let $\{\alpha_i\}$, $\alpha_i \geq 0$ be a sequence of rational numbers, $\sum_{i=1}^{\infty} \alpha_i = \alpha$. Then there exists a set of mutually orthogonal projections $\{E_i\}$ such that $E_i \leq E$, $E_i \in M$, $D_M(E_i) = \alpha_i$. To show this, suppose $E_1, \dots, E_{i-1}$ have been chosen with these properties. Then $D_M\left(E - \sum_{j=1}^{i-1} E_j\right) = \sum_{j=i}^{\infty} \alpha_j \geq \alpha_i$. Hence an E_i can be chosen in such a way that $E_i \in M$ is $\leq E - \sum_{j=1}^{i-1} E_j$ with $D_M(E_i) = \alpha_i$. Now $D_M\left(E - \sum_{i=1}^{\infty} E_i\right) = \alpha - \sum_{i=1}^{\infty} \alpha_i = 0$ or $E = \sum_{i=1}^{\infty} E_i$. Since each E_i satisfies (+), E does too. Since E is arbitrary we have

Lemma 1.6.2. *Let K and g be as in Lemma 1.5.1, then for every $E \in M$*

$$K(Eg, g) \geq D_M(E) \geq K^{-1}(Eg, g).$$

1.7. From Lemma 1.6.2 we can conclude, using Lemma 1.4.2, that if $A \in M$ is positive definite, then

$$K(Ag, g) \geq Tr_M(A) \geq K^{-1}(Ag, g).$$

We can state our result as follows.

Theorem I. *Let M be a factor in case II_1 . Let M and M' be such that when D_M and $D_{M'}$ are normalized in such a way that the range of D_M is the interval $(0, 1)$ and $C = 1$ (cf. §1.1), then the range of $D_{M'}$ is an interval $(0, \alpha)$ with $0 < \alpha \leq \infty$, then $\alpha \geq 1$. Then to every $K > 1$, there exists a $g \in \mathfrak{H}$ such that for every positive definite $A \in M$,*

$$K(Ag, g) \geq Tr_M(A) \geq K^{-1}(Ag, g).$$

CHAPTER II. IMMEDIATE CONSEQUENCES

2.1. We are now able to show that Tr_M has the following two properties when M is in a finite case (cf. R.O., Theorem VIII); that is, when M is in a case I_n ($n = 1, 2, \dots$) or II_1 .

Property I. *For all Hermitian A and $B \in M$*

$$Tr_M(A + B) = Tr_M(A) + Tr_M(B).$$

Property II. *$Tr_M(A)$ is weakly continuous if A is subject to either of these conditions:*

- (i) *A is uniformly bounded (that is, $\|A\| \leq D$ for some fixed D);*
- (ii) *A is definite.*

These properties are obviously independent of the normalization of $D_M(E)$ and $Tr_M(A)$.

If M is in cases I_n , $n = 1, 2, \dots$, these are of course well known results about $Tr_M(A)$ (cf. [R.O.](#), p. 220). So we may assume M to be in case II_1 .

Then we have in the normalization of [R.O.](#), Theorem VIII, three possibilities for the factorization M, M' : II_1, II_∞ ; II_1, II_1 with $C \leq 1$; II_1, II_1 with $C \geq 1$. The two first ones correspond to $\alpha \geq 1$ in the normalization of [§1.1](#), and since we shall only use [Theorem I](#), from [§1.1](#), we may treat these two cases together.

We therefore consider first the conditions under which [Theorem I](#) holds: the normalization of [§1.1](#), and $\alpha \geq 1$. We obtain an equivalent statement of [Theorem I](#) with $A \in M$, Hermitian and not necessarily definite.

Let K, g be as in [Theorem I](#), then $K^{-1}(g, g) = K^{-1}\|g\|^2 \leq Tr_M(1) = 1$ or $\|g\|^2 \leq K$. Furthermore if $A \in M$ is Hermitian, then $A = A_1 - A_2$, where A_1 and A_2 are both positive definite and $A_1 A_2 = 0 = A_2 A_1$. (Take $A_1 = \frac{1}{2}(A + |A|)$, $A_2 = \frac{1}{2}(|A| - A)$ (cf. [\(15\)](#), Chapter VI, or [\(19\)](#), p. 203 ff.)) Since A_1 and A_2 commute, $Tr_M(A) = Tr_M(A_1) - Tr_M(A_2)$ by [R.O.](#), Lemma 15.3.4. It is a consequence of [Theorem 1](#) that, if $\epsilon = K - 1$,

$$|Tr_M(A_1) - (A_1 g, g)| \leq \epsilon(A_1 g, g); \quad |Tr_M(A_2) - (A_2 g, g)| \leq \epsilon(A_2 g, g).$$

These with the above equation for $Tr_M(A)$ imply

$$|Tr_M(A) - (A g, g)| \leq \epsilon((A_1 g, g) + (A_2 g, g)).$$

But the bounds of A_1 and A_2 are each not more than that of A . Also $\|g\|^2 \leq 1 + \epsilon$. So if A has the bound D , we have

$$|Tr_M(A) - (A g, g)| \leq 2\epsilon(1 + \epsilon)D.$$

Let $A, B, A + B$ have the respective bounds D_1, D_2, D_3 . Substituting in [\(1\)](#) each of these operators, inasmuch as $((A + B)g, g) = (A g, g) + (B g, g)$, we get

$$|Tr_M(A + B) - Tr_M(A) - Tr_M(B)| \leq 2\epsilon(1 + \epsilon)(D_1 + D_2 + D_3),$$

which since ϵ may be taken arbitrarily small, implies [Property I](#).

We turn our attention to [Property II](#), (i). Consider the set Σ of all Hermitian $A \in M$, with a bound less than or equal to a fixed D . We will show that to every $A \in \Sigma$ and to every $\eta > 0$, there exists a weak neighborhood of A , $U(A; g; g; \eta/3)$ such that for all $B \in \Sigma$ and $B \in U(A; g; g; \eta/3)$ (i.e., $|((B - A)g, g)| < \eta/3$) we have $|Tr_M(B) - Tr_M(A)| < \eta$. Letting ϵ be such that $\epsilon(1 + \epsilon)D \leq \eta/3$, as above we can find a g such that for A and every $B \in \Sigma$, $|Tr_M(A) - (A g, g)| \leq \epsilon(1 + \epsilon)D \leq$

$\eta/3$, $|Tr_M(B) - (Bg, g)| \leq \eta/3$. In particular for $B \in U(A; g; g; \eta/3)$ (cf. above) which means that we also have $|((B - A)g, g)| < \eta/3$, these inequalities imply that $|Tr_M(A) - Tr_M(B)| < \eta$.

Thus we have shown the weak neighborhood continuity of $Tr_M(A)$ with of course a restriction. From this the sequential continuity follows immediately. (If $A_n \rightarrow A$, then the A_n are uniformly bounded and for any $\eta > 0$, almost all the A_n must be in the above given neighborhood.) In this situation the two kinds of continuity are equivalent (cf. (18), pp. 383–384), but we will use only the sequential.

$Tr_M(A)$ is also weakly continuous when considered only for the positive definite $A \in M$. For suppose A is such and an $\epsilon > 0$ is given. Let the K of Theorem I be chosen in such a way that $1 < K \leq 2$, and $(K - 1)Tr_M(A) < \epsilon/5$, and let g correspond to this K as there. Let η be chosen in such a way that $\eta \leq \epsilon/5$. Then if a positive definite $B \in M$ is in $U(A; g; g; \eta)$, we have $|Tr_M(A) - Tr_M(B)| < \epsilon$ (because $|((B - A)g, g)| < \eta \leq \epsilon/5$, $|Tr_M(A) - (Ag, g)| \leq (K - 1)Tr_M(A) < \epsilon/5$, and, in addition, $|Tr_M(B) - (Bg, g)| \leq (K - 1)(Bg, g) \leq (K - 1)((Ag, g) + \eta) \leq (K - 1)(Tr_M(A) + \epsilon/5 + \eta) \leq 3\epsilon/5$).

Thus we have settled the factorizations II_1, II_∞ and II_1, II_1 with $C \leq 1$. But we still must consider the case II_1, II_1 with $C > 1$. Let m be an integer such that $C/m \leq 1$, E_m an m -dimensional unitary space. Form $E_m \otimes \mathfrak{H}$ (cf. R.O., Theorem I). Let N be the ring of operators on E_m , N' the corresponding set of operators on $E_m \otimes \mathfrak{H}$ (cf. R.O., Lemma 2.3.1 and Lemma 2.3.2), B the set of all operators in $\mathfrak{H}$, $B^{(2)}$ the corresponding set in $E_m \otimes \mathfrak{H}$.

Under the above correspondence of B and $B^{(2)}$, $M \sim M^{(2)}$, $M' \sim M'^{(2)}$. The first and hence each of the others is a full ring isomorphism (R.O., Lemma 2.3.4 and (22), §4). Thus $M^{(2)}$ and $M'^{(2)}$ are in case II_1 (R.O., Theorem IX). Now let $\phi_1, \dots, \phi_m$ be a complete orthonormal set in E_m and let us think of the set up of R.O., §2.4. By R.O., Lemma 2.4.5, we have $M^{(2)'} = R(N^{(1)}, M'^{(2)})$ and by R.O., Lemma 11.5.2, $R(N^{(1)}, M'^{(2)})$ is in case II_1 since as we have mentioned before, $M'^{(2)}$ is in this case.

Thus the coupled factorization, $M^{(2)}, M^{(2)'}$, is in case II_1, II_1 and $M^{(2)}$ is fully ring isomorphic to M . Furthermore such an isomorphism preserves D_M in the standard normalization (R.O., Theorem IX) and hence the trace. Now suppose we have shown that for $M^{(2)}, M^{(2)'}$ we have $C^{(2)} \leq 1$. Then $M^{(2)}$ has Properties I and II, by the above and M must have them to be the isomorphism. (As m is finite, even weak operator topology is conserved under the mapping $B \sim B^{(2)}$.)

Therefore it remains to show that $C^{(2)} \leq 1$. Consider the closed linear manifold, $(\phi_1 \otimes f; f \in \mathfrak{H}) = (\text{say}) \mathfrak{M}$, in $E_m \otimes \mathfrak{H}$. The correspondence of $\mathfrak{M}$ and $\mathfrak{H}$ given by $\phi_1 \otimes$

$f \sim f$ is unitary. Under it, $M_{\mathfrak{M}}^{(2)}$ and $R(N^{(1)}, M'^{(2)})_{\mathfrak{M}}$ (cf. [R.O.](#), Definition 11.3.1) are unitarily equivalent to M and M' (cf. [R.O.](#), §2.4). Thus the C for M, M' is the same as that of $M_{\mathfrak{M}}^{(2)}$ and $R(N^{(1)}, M'^{(2)})_{\mathfrak{M}}$. Then from the proof of Lemma 11.4.3, we can conclude that $C = (D_{M^{(2)}}(\mathfrak{S})/D_{M^{(2)},(\mathfrak{M})})C^{(2)} = mC^{(2)}$ or $C^{(2)} = C/m \leq 1$.

2.2. Previously $Tr_M(A)$ has only been considered for Hermitian $A \in M$. We now define it for all $A \in M$.

Definition 2.2.1. *If $A \in M$, then $A = B + iC$, where $B = \frac{1}{2}(A + A^*)$, $C = -\frac{1}{2}i(A - A^*)$ are Hermitian, and we define $Tr_M(A) = Tr_M(B) + iTr_M(C)$.*

$Tr_M(A)$ has the following property.

Property III. (*) $Tr_M(A)$ agrees with the previous definition of $Tr_M(A)$ if A is Hermitian.

(**) For uniformly bounded or for definite A 's, $Tr_M(A)$ is weakly continuous.

- (i) $Tr_M(1) = 1$;
- (ii) $Tr_M(\rho A) = \rho Tr_M(A)$, ρ complex;
- (iii) $Tr_M(A + B) = Tr_M(A) + Tr_M(B)$;
- (iv) $Tr_M(A) \geq 0$, if A is definite;
- (iv)' If A is definite and $\neq 0$, $Tr_M(A) > 0$;
- (v) $Tr_M(A^*) = \overline{Tr_M(A)}$;
- (vi) $Tr_M(AB) = Tr_M(BA)$;
- (vi)' $Tr_M(U^{-1}AU) = Tr_M(A)$, if U is unitary.

Now, (*) and (v) are immediate consequences of the definition. (i), (iv), and (iv)' follow from (*) and [R.O.](#), Theorem XIII. A^* is a weakly continuous additive function of A . Thus B and C are too, and we may infer (iii) and (**) from [Properties I and II](#) respectively. To show (ii) we first note that if A is Hermitian by [Definition 2.2.1](#), $Tr_M(iA) = iTr_M(A)$ and if a is real by (*) $Tr_M(aA) = aTr_M(A)$. Then (ii) can be shown by a calculation involving (iii).

To show (vi) suppose A and $B \in M$ are Hermitian. Let $C = A + iB$, $C^* = A - iB$. Then $Tr_M(CC^*) = Tr_M(C^*C)$ (by (*), cf. proof of [Lemma 1.4.3](#), equation (β)). Substituting for C , expanding the products according to the distributive law and collecting terms, we get $2i(Tr_M(AB) - Tr_M(BA)) = 0$ or $Tr_M(AB) = Tr_M(BA)$. From this we can show (vi) in general by using [Definition 2.2.1](#), (ii) and (iii). (vi)' follows from (vi) by letting $A = U^{-1}A$, $B = U$ and using the associative law.

Property IV. (i)–(iv), (v), (vi)' of [Property III](#) characterize $Tr_M(A)$ uniquely.

Let $Tr'(A)$ have the listed properties. Then if A is Hermitian, $Tr'(A)$ is real by (v). This and (i)–(iv), (vi)' imply that for $A \in M$ and Hermitian $Tr'(A) = Tr_M(A)$ ([R.O.](#), Theorem XIII). By (ii) and (iii), we then have that for an arbitrary $C \in M$,

$C = A + iB$, A and B Hermitian, $Tr'(C) = Tr'(A + iB) = Tr'(A) + Tr'(iB) = Tr'(A) + iTr'(B) = Tr_M(A) + iTr_M(B) = Tr_M(A + iB) = Tr_M(C)$ or $Tr'(C) = Tr_M(C)$.

CHAPTER III. THE EXACT FORM OF $Tr_M(A)$ (FOR $\alpha \geq 1$)

3.1. We again assume M in case II_1 . Some lemmas about families of projections and spectral forms in M are needed. Of these the two last ones have some interest of their own.

Lemma 3.1.1. *For any two projections $E, F \in M$ with $E \leq F$, and any $\alpha \geq D_M(E)$, $\leq D_M(F)$, a projection $G \in M$ with $E \leq G \leq F$, $D_M(G) = \alpha$ exists.*

$F - E$ is a projection, and $0 \leq \alpha - D_M(E) \leq D_M(F) - D_M(E) = D_M(F - E) \leq 1$. Thus a projection $G' \in M$ with $D_M(G') = \alpha - D_M(E)$ exists (because M is in case II_1). So $D_M(G') \leq D_M(F - E)$ and therefore a projection $G'' \in M$ with $G'' \leq F - E$, $D_M(G'') = D_M(G') = \alpha - D_M(E)$ exists. G'' is thus orthogonal to E , and therefore $G'' + E$ is a projection and $D_M(G'' + E) = D_M(G'') + D_M(E) = \alpha$. Besides, $G'' + E \geq E$ and $G'' + E \leq (F - E) + E = F$. So $G = G'' + E$ has the desired properties.

Lemma 3.1.2. *For any two projections $E, F \in M$ with $E \leq F$ a family of projections $G(\alpha) \in M$ defined for all α with $D_M(E) \leq \alpha \leq D_M(F)$ exists, which possesses the following properties:*

- (i) $G(\alpha) = E$ or F if $\alpha = D_M(E)$ or $D_M(F)$ respectively.
- (ii) $\alpha \leq \beta$ implies $G(\alpha) \leq G(\beta)$.
- (iii) $D_M(G(\alpha)) = \alpha$.

Choose a sequence $\rho_1, \rho_2, \dots$ which lies and is dense in the interval $D_M(E) \leq \alpha \leq D_M(F)$, with $\rho_1 = D_M(E)$, $\rho_2 = D_M(F)$.

We will now define a sequence of projections $G(\rho_1), G(\rho_2), \dots \in M$, so that (i)–(iii) hold for $\alpha = \rho_1, \rho_2, \dots$. Put first $G(\rho_1) = E$, $G(\rho_2) = F$; then (i)–(iii) hold for $\alpha = \rho_1, \rho_2$. Assume now that for a $j = 3, 4, \dots$ the $G(\rho_1), \dots, G(\rho_{j-1})$ are already defined, so that (i)–(iii) holds for $\alpha = \rho_1, \dots, \rho_{j-1}$, we will now define $G(\rho_j)$ without violating (i)–(iii).

Consider the $\rho_i, i = 1, \dots, j-1$, with $\rho_i \leq \rho_j$ (they exist: $\rho_1 = D_M(E) \leq \rho_j$); let $\rho_{i'}$ be the greatest one. Consider the $\rho_i, i = 1, \dots, j-1$, with $\rho_i \geq \rho_j$ (they exist: $\rho_2 = D_M(F) \geq \rho_j$); let $\rho_{i''}$ be the smallest one. Thus $\rho_{i'} \leq \rho_{i''}$, and so $G(\rho_{i'}) \leq G(\rho_{i''})$. Besides $D_M(G(\rho_{i'})) = \rho_{i'} \leq \rho_j \leq \rho_{i''} = D_M(G(\rho_{i''}))$. So **Lemma 3.1.1** can be applied to $E = G(\rho_{i'})$, $F = G(\rho_{i''})$, $\alpha = \rho_j$, and we define $G(\rho_j) = G$.

Thus $G(\rho_{i'}) \leq G(\rho_j) \leq G(\rho_{i''})$, therefore $\rho_i \leq \rho_j, i = 1, \dots, j-1$, implies $\rho_i \leq \rho_{i'}$, $G(\rho_i) \leq G(\rho_{i'}) \leq G(\rho_j)$ and $\rho_i \geq \rho_j, i = 1, \dots, j-1$, implies $\rho_i \geq \rho_{i''}$, $G(\rho_i) \geq G(\rho_{i''}) \geq G(\rho_j)$. Besides $D_M(G(\rho_j)) = \rho_j$. So (i)–(iii) hold for $\alpha = \rho_1, \dots, \rho_{j-1}, \rho_j$, too.

Thus all $G(\rho_1), G(\rho_2), \dots$ are defined. Owing to (ii), $\lim_{\rho_i \rightarrow \alpha, \rho_i \geq \alpha} G(\rho_i)$ exists for all α with $D_M(E) = \rho_1 \leq \alpha \leq \rho_2 = D_M(F)$. If α is equal to a ρ_j , then this limit is meant to denote the value at ρ_j . So we can extend the definition of $G(\alpha)$ to all above α by defining

$$G(\alpha) = \lim_{\rho_i \rightarrow \alpha, \rho_i \geq \alpha} G(\rho_i).$$

Now (i) remains true, and (ii), (iii) extend by continuity to all above α .

Lemma 3.1.3. *Let $A \in M$ be a definite operator with bound ≤ 1 . Then there exists a monotonous non-decreasing and right semi-continuous function $\phi(\alpha)$ for $0 \leq \alpha \leq 1$ with $0 \leq \phi(\alpha) \leq 1$ and a resolution of identity $E(\alpha) \in M$ (cf. (16), p. 92, or R.O., Definition 15.1.1) with the following properties:*

- (i) $E(0) = 0, E(1) = 1$,
- (ii) $D_M(E(\alpha)) = \alpha$,
- (iii) $(Af, g) = \int_0^1 \phi(\alpha) d(E(\alpha)f, g)$ for any f, g , or symbolically,

$$A = \int_0^1 \phi(\alpha) dE(\alpha).$$

Let $F(\lambda)$ be the resolution of identity corresponding to A . As A is definite, so $F(0) = 0$; as the bound of A is ≤ 1 , so $F(1) = 1$.

Consider now the two following functions:

$$\psi(\lambda) = D_M(F(\lambda)), \quad \phi(\alpha) = \underset{\substack{0 \leq \lambda \leq 1 \\ \psi(\lambda) \leq \alpha}}{\text{l. u. b. } \lambda}.$$

$\psi(\lambda)$ is monotonous non-decreasing and right semi-continuous in $0 \leq \lambda \leq 1$, because $F(\lambda)$ is a resolution of identity, and besides $\psi(0) = 0, \psi(1) = 1$. Thus $\phi(\alpha)$ is monotonous non-decreasing and right semi-continuous in $0 \leq \alpha \leq 1$, and its values are all in $0 \leq \lambda \leq 1$. One verifies these facts, as well as those which follow, easily; besides, their discussion may be found in (19) (numbers in parentheses refer to the bibliography in R.O., pp. 125–126) on p. 193. (Our $\psi(\lambda)$, $\phi(\alpha)$ corresponds to the $\phi(a), \psi(b)$ there, thus they are both “ m -functions,” and $\phi(\alpha)$ is the “measure function” of $\psi(\lambda)$.) The relation between $\phi(\alpha)$ and $\psi(\lambda)$ is symmetric (cf. loc. cit., $\psi(\lambda)$ is the “measure function” of $\phi(\alpha)$, we must define for the empty set in $0 \leq \lambda \leq 1$ the l.u.b. 0); that is,

$$\psi(\lambda) = \underset{0 \leq \alpha \leq 1}{\text{l. u. b. } (\phi(\alpha) \leq \lambda)}.$$

Finally (cf. loc. cit.)

$$(*) \quad \psi(\phi(\psi(\lambda))) = \psi(\lambda).$$

That is, $D_M(F(\phi(\psi(\lambda)))) = D_M(F(\lambda))$. But $F(\phi(\psi(\lambda))) \geq F(\lambda)$, and therefore this implies

$$(**) \quad F(\phi(\psi(\lambda))) = F(\lambda).$$

If $\phi(\alpha) < 1$, then choose μ with $\phi(\alpha) < \mu \leq 1$, and let $\mu \rightarrow \phi(\alpha)$. Under these conditions the definition of $\phi(\alpha)$ implies $D_M(F(\mu)) > \alpha$, so $D_M(F(\phi(\alpha))) \geq \alpha$. For $\phi(\alpha) = 1$ this holds, too: $D_M(F(\phi(\alpha))) = D_M(F(1)) = D_M(1) = 1 \geq \alpha$. On the other hand, if $\phi(\alpha) > 0$, then a sequence of μ with $0 \leq \mu \leq \phi(\alpha)$, $\mu \rightarrow \phi(\alpha)$ and $D_M(F(\mu)) \leq \alpha$ exists, so $D_M(F(\phi(\alpha) - 0)) \leq \alpha$. This holds for $\phi(0) = 0$, too, if we identify $F(0 - 0)$ with $F(0) = 0$: $D_M(F(0 - 0)) = D_M(0) = 0 \leq \alpha$. So we have

$$(\#) \quad D_M(F(\phi(\alpha) - 0)) \leq \alpha \leq D_M(F(\phi(\alpha))).$$

Form now for every $0 \leq \lambda \leq 1$, $E = F(\lambda - 0)$, $F = F(\lambda)$ and apply [Lemma 3.1.2](#). A family of projections $G(\alpha) \in M$, $D_M(F(\lambda - 0)) \leq \alpha \leq D_M(F(\lambda))$, results. Choose for every $0 \leq \lambda \leq 1$ such a family, and hold it fixed: $G(\lambda, \alpha)$. (If $F(\lambda - 0) = F(\lambda)$, then necessarily $\alpha = D_M(F(\lambda))$, and $G(\lambda, \alpha) = F(\lambda)$. So only the λ with $F(\lambda - 0) \neq F(\lambda)$, the point-proper values of A , are important. Their set is finite or enumerably infinite.)

Now define for every $0 \leq \alpha \leq 1$

$$(\S) \quad E(\alpha) = G(\phi(\alpha), \alpha)$$

(this expression has a meaning owing to [\(\#\)](#)).

We have now by definition $D_M(E(\alpha)) = \alpha$, so condition (ii) is satisfied. For $\alpha = 0, 1$, this gives $E(0) = 0$, $E(1) = 1$, so condition (i) is satisfied, too. If $\alpha \leq \beta$, then $\phi(\alpha) \leq \phi(\beta)$. The relation $\phi(\alpha) = \phi(\beta)$ gives

$$E(\alpha) = G(\phi(\alpha), \alpha) = G(\phi(\beta), \alpha) \leq G(\phi(\beta), \beta) = E(\beta),$$

while $\phi(\alpha) < \phi(\beta)$ gives

$$E(\alpha) = G(\phi(\alpha), \alpha) \leq F(\phi(\alpha)) \leq F(\phi(\beta) - 0) \leq G(\phi(\beta), \beta) = E(\beta).$$

So we see that

$$(\cdot) \quad \alpha \leq \beta \quad \text{implies} \quad E(\alpha) \leq E(\beta).$$

Furthermore $\beta > \alpha$ gives

$$D_M(E(\beta) - E(\alpha)) = D_M(E(\beta)) - D_M(E(\alpha)) = \beta - \alpha,$$

so $\lim_{\substack{\beta \rightarrow \alpha \\ \beta > \alpha}} D_M(E(\beta) - E(\alpha)) = 0$. But

$$\lim_{\substack{\beta \rightarrow \alpha \\ \beta > \alpha}} D_M(E(\beta) - E(\alpha)) = D_M \left(\lim_{\substack{\beta \rightarrow \alpha \\ \beta > \alpha}} E(\beta) - E(\alpha) \right) = D_M(E(\alpha + 0) - E(\alpha)),$$

so $D_M(E(\alpha + 0) - E(\alpha)) = 0$, $E(\alpha + 0) = E(\alpha)$, or

$$(\cdots) \quad \lim_{\substack{\beta \rightarrow \alpha \\ \beta > \alpha}} E(\beta) = E(\alpha).$$

(i), $(\cdot)$, $(\cdots)$ state that $E(\alpha)$ is a resolution of unity.

It remains for us to prove (iii).

Consider the expression $\int_0^1 \phi(\alpha) d(\|E(\alpha)f\|^2)$. $\phi(\alpha)$ and $\|E(\alpha)f\|^2$ are both monotonous non-decreasing functions of α ; if α increases from 0 to 1, then these functions increase from $\phi(0)$ to $\phi(1) = 1$ and from 0 to $\|f\|^2$ respectively. Thus our integral is, by partial integration, equal to $\|f\|^2 - \int_0^1 \|E(\alpha)f\|^2 d\phi(\alpha)$. We may now introduce a new variable of integration: $\lambda = \phi(\alpha)$. Then we have (cf. loc. cit., p. 198) $\int_0^1 \|E(\alpha)f\|^2 d\phi(\alpha) = \int_0^1 \|E(\psi(\lambda))f\|^2 d\lambda$. But by (§), $E(\psi(\lambda)) = G(\phi(\psi(\lambda)), \psi(\lambda))$, and by (*), $\psi(\phi(\psi(\lambda))) = D_M(F(\phi(\psi(\lambda))))$. So the definition of $G(\mu, \alpha)$ gives $G(\phi(\psi(\lambda)), \psi(\lambda)) = F(\phi(\psi(\lambda)))$. By (**) this is $F(\lambda)$, so we have $E(\psi(\lambda)) = F(\lambda)$. Thus our original expression is equal to $\|f\|^2 - \int_0^1 \|F(\lambda)f\|^2 d\lambda$, and this becomes by another partial integration $\int_0^1 \lambda d\|F(\lambda)f\|^2$, that is, (Af, f) . Thus we have proved:

$$(Af, f) = \int_0^1 \phi(\alpha) d\|E(\alpha)f\|^2.$$

Replacing herein f by $f \pm g/2$, and subtracting, gives the real part of (iii). Now replacing f, g by if, g gives the imaginary part of (iii). Thus the proof is completed.

Finally we need the following lemma.

Lemma 3.1.4. *Let $E(\alpha) \in M$ be a resolution of unity in $a \leq \alpha \leq b$, $E(a) = 0$, $E(b) = 1$. Then a bounded Hermitian operator $B \in M$ with $(Bf, g) = \int_a^b \lambda d(E(\lambda)f, g)$, symbolically $B = \int_a^b \lambda dE(\lambda)$, exists. For every bounded Baire function $\psi(\alpha)$ an operator $C = \psi(B) \in M$ with $(Cf, g) = \int_a^b \psi(\lambda) d(E(\lambda)f, g)$, symbolically $C = \int_a^b \psi(\lambda) dE(\lambda)$ exists. If $0 \leq \psi(\lambda) \leq 1$, then C is Hermitian, definite, and of bound 1. For all $\psi(\lambda)$*

$$Tr_M(C) = \int_a^b \psi(\lambda) dD_M(E(\lambda)).$$

The existence of the bounded B is a well known fact about spectral forms, as all $E(\lambda) \in M$ so $B \in M$ (see (18), p. 389, Theorem 1). $C = \psi(B)$ exists and is bounded ((19), p. 205), and $B \in M$ implies $C \in M$ ((19), p. 213, Theorem 6). $0 \leq \psi(\alpha) \leq 1$ implies the Hermitian, definite character of C and $1 - C$ ((19), p. 205) (C Hermitian: by Property b; C and $1 - C$ definite: by Property e), with $F(x) = \psi(x)$ or $1 - \psi(x)$ and $G(x) = H(x) = (F(x))^{1/2}$. Thus the bound of C is ≤ 1 . (Alternatively (16), p. 113, Theorem 4*, could be used.)

It remains to prove the final formula for the trace. If it holds for $\psi_1(\lambda), \psi_2(\lambda), \dots$, and the $\psi_n(\lambda), n = 1, 2, \dots$ are uniformly bounded and everywhere convergent in $a \leq \lambda \leq b$, then it holds for $\psi(\lambda) = \lim_{n \rightarrow \infty} \psi_n(\lambda)$ too: $\psi(B) = \text{strong } \lim_{n \rightarrow \infty} \psi_n(B)$, the $\psi_n(B)$ being uniformly bounded, by (19), p. 205, Property h), and Tr_M is continuous for strong (even for weak) convergence by Property III, (**). Thus our relation holds for all bounded Baire functions $\psi(\lambda)$ if it holds for all continuous functions $\psi(\lambda)$. And it holds for these, if it holds for all intervalwise constant functions $\psi(\lambda)$. But these are linear aggregates of functions $\psi_{c,d}(\lambda) = 1$ for $c < \lambda \leq d$ = 0 otherwise, where $a \leq c \leq d \leq b$. So it suffices to consider these. Now clearly

$$\begin{aligned}\psi_{c,d}(B) &= E(d) - E(c), \\ Tr_M(\psi_{c,d}(B)) &= D_M(E(d) - E(c)) = D_M(E(d)) - D_M(E(c)),\end{aligned}$$

and

$$\int_a^b \psi_{c,d}(\alpha) dD_M(E(\alpha)) = D_M(E(d)) - D_M(E(c)),$$

completing the proof.

3.2. We return now to the normalization of §1.1, and assume that an M in case Π_1 is given with $\alpha \geq 1$.

Consider a g and K which are related as in Theorem I. Our objective is to prove Theorem II below, but we begin by considering these two maximum problems.

(A) For a given λ with $0 \leq \lambda \leq 1$ consider all projections $F \in M$ with $D_M(F) = \lambda$, and the corresponding values of (Fg, g) . Prove that these (Fg, g) possess a maximum m_a which they assume for a certain $F = F_0$ from the above class.

(B) For a given λ with $0 \leq \lambda \leq 1$ consider all definite operators $B \in M$ of bound ≤ 1 with $Tr_M(B) = \lambda$, and the corresponding values of (Bg, g) . Prove that these (Bg, g) possess a maximum m_b which they assume for a certain $B = B_0$ from the above class.

Lemma 3.2.1. *Problem (B) possesses a solution $B = B_0$, the maximum m_b fulfills $\lambda K^{-1} \leq m_b \leq \lambda K$.*

For the B of the class described in Problem (B) we have by [Theorem I](#), $\lambda K^{-1} \leq (Bg, g) \leq \lambda K$. So these (Bg, g) possess a l.u.b. m'_b and $\lambda K^{-1} \leq m'_b \leq \lambda K$.

Select now a sequence $B_1, B_2, \dots$ from this class, so that $\lim_{n \rightarrow \infty} (B_n g, g) = m'_b$. As the $B_1, B_2, \dots$ are uniformly bounded, there exists a subsequence $B_{n_1}, B_{n_2}, \dots$ ($n_1 < n_2 < \dots$) such that $B_0 = \text{weak } \lim_{i \rightarrow \infty} B_{n_i}$ exists. As all B_{n_i} are in M , definite, and of bound ≤ 1 , the same is true for B_0 . As all $\text{Tr}_M(B_{n_i}) = \lambda$, so Property II or III, (**), gives $\text{Tr}_M(B_0) = \lambda$. Finally $(B_0 g, g) = \lim_{i \rightarrow \infty} (B_{n_i} g, g) = \lim_{n \rightarrow \infty} (B_n g, g) = m'_b$.

Thus B_0 belongs to our class, and $(B_0 g, g) = m'_b$. Therefore the l.u.b. m'_b is a maximum: $m'_b = m_b$, and $\lambda K^{-1} \leq m_b \leq \lambda K$. So the proof is completed.

Lemma 3.2.2. *The B_0 of [Lemma 3.2.1](#) can be chosen as a projection.*

Consider the B_0 of [Lemma 3.2.1](#). By [Lemmas 3.1.3](#) and [3.1.4](#) we have $B_0 = \phi(B)$, where $B = \int_0^1 \alpha dE(\alpha)$ (symbolically), $E(\alpha)$ being a resolution of unity with $E(\alpha) \in M$, $D_M(E(\alpha)) = \alpha$, and $\phi(\alpha)$ a monotonous non-decreasing and right semi-continuous function.

Consider any Baire function $\psi(\alpha)$, $0 \leq \alpha \leq 1$, with $0 \leq \psi(\alpha) \leq 1$. Form $\psi(B) = \int_0^1 \psi(\alpha) dE(\alpha)$ (symbolically), using [Lemma 3.1.4](#). Then $\psi(B)$ is in M , definite, and of bound ≤ 1 , and

$$\text{Tr}_M(\psi(B)) = \int_0^1 \psi(\alpha) dD_M(E(\alpha)) = \int_0^1 \psi(\alpha) d\alpha.$$

Thus $\psi(B)$ belongs to the class of Problem (B) if and only if $\int_0^1 \psi(\alpha) d\alpha = \lambda$. Besides $(\psi(B)g, g) = \int_0^1 \psi(\alpha) d(\|E(\alpha)g\|^2)$.

Apply this to $\psi(\alpha) = \phi(\alpha)$, $\psi(B) = \phi(B) = B_0$. We get $\int_0^1 \phi(\alpha) d\alpha = \lambda$, $\int_0^1 \phi(\alpha) d(\|E(\alpha)g\|^2) = m_b$. And the maximum property of B_0 gives the following. For every Baire function $\psi(\alpha)$, $0 \leq \alpha \leq 1$, with $0 \leq \psi(\alpha) \leq 1$ the equation $\int_0^1 \psi(\alpha) d\alpha = \lambda$ implies $\int_0^1 \psi(\alpha) d\|E(\alpha)g\|^2 \leq m_b$.

If we had $\phi(1 - \lambda) = 0$, then the monotony of $\phi(\alpha)$ would give $\phi(\alpha) = 0$ for $0 \leq \alpha \leq 1 - \lambda$, and the right semi-continuity $\phi(\alpha) \leq \frac{1}{2}$ for $1 - \lambda < \alpha \leq 1 - \lambda + \epsilon'$ for a suitable $\epsilon' > 0$. Besides $\phi(\alpha)$ is always ≤ 1 . Thus

$$\int_0^1 \phi(\alpha) d\alpha \leq 0(1 - \lambda) + \frac{1}{2}\epsilon' + (\lambda - \epsilon')1 = \lambda - \frac{1}{2}\epsilon' < \lambda,$$

contradicting $\int_0^1 \phi(\alpha) d\alpha = \lambda$. Therefore $\phi(1 - \lambda) > 0$.

If we had $\phi(1 - \lambda - 0) = 1$, then the monotony of $\phi(\alpha)$ would give $\phi(\alpha) = 1$ for $1 - \lambda \leq \alpha \leq 1$ and the definition of $\phi(1 - \lambda - 0)$, $\phi(\alpha) \geq \frac{1}{2}$ for $1 - \lambda - \epsilon' \leq \alpha < 1 - \lambda$,

for a suitable $\epsilon' > 0$. Besides $\phi(\alpha)$ is always ≥ 0 . Thus

$$\int_0^1 \phi(\alpha) d\alpha \geq (1 - \lambda - \epsilon') \cdot 0 + \frac{\epsilon'}{2} + \lambda \cdot 1 = \lambda + \frac{\epsilon'}{2} > \lambda,$$

contradicting $\int_0^1 \phi(\alpha) d\alpha = \lambda$. Therefore $\phi(1 - \lambda - 0) < 1$.

So we can choose a δ , $0 < \delta < 1$, with $\phi(1 - \lambda) > \delta$, $\phi(1 - \lambda - 0) < 1 - \delta$. Thus $1 - \lambda \leq \alpha \leq 1$ implies $(\phi(\alpha) - \delta)/(1 - \delta) \geq (\phi(1 - \lambda) - \delta)/(1 - \delta) \geq 0$ and $\leq (1 - \delta)/(1 - \delta) = 1$, and $0 \leq \alpha \leq 1 - \lambda$ implies $\phi(\alpha)/(1 - \delta) \geq 0$ and $\leq (1 - \delta)/(1 - \delta) = 1$. Now put $\phi_1(\alpha) = 1$ for $1 - \lambda \leq \alpha \leq 1$ and $\phi_1(\alpha) = 0$ for $0 \leq \alpha \leq 1 - \lambda$. Then we have $0 \leq \phi_1(\alpha) \leq 1$ for all $0 \leq \alpha \leq 1$; and also, for $\phi_2(\alpha) = (\phi(\alpha) - \delta\phi_1(\alpha))/(1 - \delta)$, $0 \leq \phi_2(\alpha) \leq 1$ for all $0 \leq \alpha \leq 1$ (we proved this above separately for $1 - \lambda \leq \alpha \leq 1$ and for $0 \leq \alpha \leq 1 - \lambda$). Besides

$$(*) \quad \phi(\alpha) = \delta\phi_1(\alpha) + (1 - \delta)\phi_2(\alpha).$$

Now clearly $\int_0^1 \phi_1(\alpha) d\alpha = \lambda$. This and $\int_0^1 \phi(\alpha) d\alpha = \lambda$ and $(*)$ give $\int_0^1 \phi_2(\alpha) d\alpha = \lambda$. Therefore $\int_0^1 \phi_1(\alpha) d\|E(\alpha)g\|^2 \leq m_b$, $\int_0^1 \phi_2(\alpha) d\|E(\alpha)g\|^2 \leq m_b$. But $(*)$ gives

$$\begin{aligned} & \delta \int_0^1 \phi_1(\alpha) d\|E(\alpha)g\|^2 + (1 - \delta) \int_0^1 \phi_2(\alpha) d\|E(\alpha)g\|^2 \\ &= \int_0^1 \phi(\alpha) d\|E(\alpha)g\|^2 = m_b. \end{aligned}$$

Therefore necessarily $\int_0^1 \phi_1(\alpha) d\|E(\alpha)g\|^2 = m_b$. In other words, for $B_1 = \phi_1(B)$ which belongs to the class of Problem (B), we have $(B_1g, g) = m_b$.

Thus $B_1 = \phi_1(B)$ is a solution of Problem (B), too. But one verifies easily that $B_1 = \phi_1(B) = 1 - E(\lambda)$, and so B_1 is a projection.

Therefore replacement of B_0 by B_1 completes the proof.

Lemma 3.2.3. *Problems (A) and (B) possess a common solution $E_0 = B_0$, and for the maxima we have $m_a = m_b$.*

The class of the E in Problem (A) is obviously a subclass of the B in Problem (B): It consists of all projections of the latter class. (This is so, because for projections E , $D_M(E) = Tr_M(E)$ (cf. [R.O.](#), Lemma 15.3.1).) Now [Lemma 3.2.2](#) states, that a solution of Problem (B) exists, which is a projection and thus belongs to the class of Problem (A). Therefore it is a solution of Problem (A) too, and $m_a = m_b$.

Problems (A) and (B) can be modified, when a projection $G = P_{\mathfrak{N}} \in M$ is given, in the following sense: Replace $\mathfrak{H}, M$ by $\mathfrak{N}, M_{(\mathfrak{N})}$ (cf. [R.O.](#), §11.3). Then

$D_M(E)$ must be replaced by $D_M(E)/D_M(G)$, in order to conserve the standard normalization and so the normalization of §1.1 requires replacing of $D_{M'}(E')$ by $D_{M'}(E')/D_M(G)$. This α is replaced by $\alpha/D_M(G)$ and so $\alpha \geq 1$ remains true. Lemma 3.2.3 is modified, in so far as λ is replaced by $\lambda/D_M(G)$ and M in Problems (A) and (B) is to be replaced by $M_{(\mathfrak{N})}$. This means that projections $F \in M$ must be $\leq G$, and operators $B \in M$ must fulfill $BG = GB = B$.

Combining this and the corresponding changes in Lemma 3.2.1 (observe that $(B_0g, g) = (E_0g, g) = \|E_0g\|^2$), we have

Corollary. *Let a projection $G \in M$ be given. Replace Problems (A) and (B) by Problems (A_G) and (B_G) , which arise by imposing these further restrictions: In (A_G) the projections $F \in M$ are $F \leq G$. In (B_G) the operators $B \in M$ fulfill $BG = GB = B$. Assume $0 \leq \lambda \leq D_M(G)$. Then Problems (A_G) and (B_G) possess a common solution $E_0 = B_0$, and we have for the maxima $K^{-1}\|E_0g\|^2 \leq m_a = m_b \leq K\|E_0g\|^2$.*

3.3. We hold g fixed, and make the following

Definition 3.3.1. *Let E, G be two projections in M , $E \leq G$. We say that E reduces g with respect to G if for every $A \in M$ with $AG = GA = A$*

$$(AEG, g) = (EAg, g).$$

If G may be taken $= 1$, we say that E reduces g .

Lemma 3.3.1. *Let G be a projection in M , E_0 a solution of Problem (A_G) (cf. the corollary to Lemma 3.2.3). Then E_0 reduces g with respect to G .*

If U is unitary, in M , and commutes with G , then $U^{-1}E_0U$ is a projection in M , it is $\leq U^{-1}GU = G$, and $D_M(U^{-1}E_0U) = D_M(E_0) = \lambda$. So $(U^{-1}E_0Ug, g) \leq (E_0g, g)$. Now $(U^{-1}E_0Ug, g) = (U^*E_0Ug, g) = (E_0Ug, Ug)$. So $(E_0g, g) - (E_0Ug, Ug) \geq 0$.

Let A be Hermitian, in M , and commute with G . Put for some $\epsilon \geq 0$, $\phi_\epsilon(\alpha) = (1 + i\epsilon\alpha)/(1 - i\epsilon\alpha)$. As $|\phi_\epsilon(\alpha)| = 1$, so $U_\epsilon = \phi_\epsilon(A)$ is unitary, and it is in M and commutes with G along with A . Further $\phi_\epsilon(\alpha) = 1 + 2i\epsilon\alpha + \epsilon^2\psi_\epsilon(\alpha)$, where $\psi_\epsilon(\alpha) = -\alpha^2/(1 - i\epsilon\alpha)$. So $|\psi_\epsilon(\alpha)| \leq D^2$ if $|\alpha| \leq D$, and therefore $\|\psi_\epsilon(A)\| \leq \|A\|^2$ (cf., for instance, (16), p. 113, Theorem 4*). Now

$$\begin{aligned} 0 &\leq (E_0g, g) - (E_0U_\epsilon g, U_\epsilon g) = (E_0g, g) \\ &\quad - (E_0(1 + 2i\epsilon A + \epsilon^2\psi_\epsilon(A))g, (1 + 2i\epsilon A + \epsilon^2\psi_\epsilon(A))g) \\ &= 2\epsilon(i(AE_0 - E_0A)g, g) + O(\epsilon^2). \end{aligned}$$

So we have

$$2\epsilon i((AE_0 - E_0A)g, g) + O(\epsilon^2) \geq 0.$$

As $\epsilon \geq 0$, this necessitates $((AE_0 - E_0A)g, g) = 0$, that is,

$$(AE_0g, g) = (E_0Ag, g).$$

This equation extends from the Hermitian $A \in M$ to all $A \in M$, which commute with G . Put for such an A , $A_1 = \frac{1}{2}(A + A^*)$, $A_2 = -\frac{1}{2}i(A - A^*)$, then it holds for A_1, A_2 and so for $A = A_1 + iA_2$. Thus we have established that E_0 reduces g with respect to G .

Lemma 3.3.2. *If E reduces g relatively to G and G reduces g , then E reduces g .*

We must show that for every $A \in M$, $(AEg, g) = (EAg, g)$. Now

$$\begin{aligned} (EAg, g) &= (GEAg, g) = (G(EA)g, g) = ((EA)Gg, g) = (EGAGg, g) \\ &= (GAGEg, g) = (GAEG, g) = ((AE)Gg, g) = (AEg, g). \end{aligned}$$

Lemma 3.3.3. *If $\{E_i\}$ is a sequence of projections $E_i \in M$ each of which reduce g and $E_i \cdot E_j = 0$ if $i \neq j$, then $\sum_{i=1}^{\infty} E_i$ reduces g . If E_1 and E_2 reduce g and $E_1 \geq E_2$ then $E_1 - E_2$ reduces g .*

This is immediate for $A \in M$ implies A is bounded.

Lemma 3.3.4. *Let $\{E_i\}$ be a sequence with the same properties as in [Lemma 3.3.3](#). Let $E = \sum_{i=1}^{\infty} E_i$. Then for $A \in M$,*

$$(EAg, g) = (AEg, g) = (EAEg, g) = \sum_{i=1}^{\infty} (E_iAE_ig, g).$$

Also if E reduces g , $EF = 0$, $F \in M$, then $(EAFg, g) = (FAEG, g) = 0$.

We have $(EAg, g) = (E(EA)g, g) = (EAEg, g) = (E(AE)g, g) = ((AE)Eg, g) = (AEg, g)$. Hence we have $(EAg, g) = ((\sum_{i=1}^{\infty} E_i)Ag, g) = \sum_{i=1}^{\infty} (E_iAg, g) = \sum_{i=1}^{\infty} (E_iAE_ig, g)$. Also $(EAFg, g) = (E(AF)g, g) = ((AF)Eg, g) = (A(FE)g, g) = 0 = ((EF)Ag, g) = (E(FA)g, g) = ((FA)Eg, g) = (FAEG, g)$.

Lemma 3.3.5. *Let $\{E_i\}$ be a sequence of mutually orthogonal projections. If $E = \sum_{i=1}^{\infty} E_i$ and A and $E_i \in M$,*

$$Tr_M(EAE) = \sum_{i=1}^{\infty} Tr_M(E_iAE_i).$$

By Property III, (vi), $Tr_M(EAE) = Tr_M(AE \cdot E) = Tr_M(AE)$. Similarly $Tr_M(E_iAE_i) = Tr_M(AE_i)$. By Property III, (**), and (iii) $Tr_M(AE) = Tr_M(\sum_{i=1}^{\infty} AE_i) = \sum_{i=1}^{\infty} Tr_M(AE_i)$. Thus $Tr_M(EAE) = \sum_{i=1}^{\infty} Tr_M(E_iAE_i)$. Note that we have also demonstrated the convergence of $\sum_{i=1}^{\infty} Tr_M(E_iAE_i)$.

Lemma 3.3.6. Let $\{E_i\}$ be a sequence of projections each $E_i \underset{p}{\geq} \lambda$ ($\underset{p}{\leq} \lambda$) for g . Furthermore let us suppose that each E_i reduces g and $E_i \cdot E_j = 0$ if $i \neq j$. Then $E = \sum_{i=1}^{\infty} E_i \underset{p}{\geq} \lambda$ ($\underset{p}{\leq} \lambda$).

Let A be positive definite, in M and such that $EA = AE = A$. Then by Lemma 3.3.4

$$(Ag, g) = (EAg, g) = \sum_{i=1}^{\infty} (E_i A E_i g, g).$$

Now $E_i A E_i$ is positive definite and is unchanged by left- or right-multiplication with E_i . Hence Lemma 1.4.2 yields $(E_i A E_i g, g) \geq \lambda \text{Tr}_M(E_i A E_i)$. Thus by Lemma 3.2.6

$$(Ag, g) = \sum_{i=1}^{\infty} (E_i A E_i g, g) \geq \lambda \sum_{i=1}^{\infty} \text{Tr}_M(E_i A E_i) = \lambda \text{Tr}_M(A).$$

Lemma 1.4.2 now implies $E \underset{p}{\geq} \lambda$.

3.4. We still suppose that g is held fixed.

Lemma 3.4.1. Let $E \neq 0$ be a projection, in M , and such that $a \underset{p}{\leq} E \underset{p}{\leq} b$. Let a λ with $0 < \lambda < D_M(E)$ be given. Then there exists a projection $E_0 \in M$ with the following properties: (i) $E_0 \leq E$; (ii) $D_M(E_0) = \lambda$; (iii) E_0 reduces g with respect to E ; and (iv) there exists a ξ_0 with $a \underset{p}{\leq} E - E_0 \underset{p}{\leq} \xi_0 \underset{p}{\leq} E_0 \underset{p}{\leq} b$.

Consider the solution E_0 of Problem (A_G) with $G = E$, in the corollary to Lemma 3.2.3. This E_0 is a projection in M and fulfills (i), (ii) by definition, and it fulfills (iii) by Lemma 3.3.1. As to (iv), both E_0 and $E - E_0$ are $\underset{p}{\geq} a$ and $\underset{p}{\leq} b$ along with E , by Lemma 1.2.3.

So we must only find a ξ with $E - E_0 \underset{p}{\leq} \xi \underset{p}{\leq} E_0$. Let Γ' be the set of all η 's, for which there exists an $F \leq E_0$ with $F < \eta$; and let Γ'' be the set of all η 's for which there exists an $F \leq E - E_0$ with $F > \eta$. Γ' and Γ'' are open intervals, and clearly every $\eta > b$ belongs to Γ' , and every $\eta < a$ belongs to Γ'' . So there exists either a ξ_0 such that every η in Γ' is $\geq \xi_0$ and every η in Γ'' is $\leq \xi_0$, or Γ' and Γ'' have a common element ξ_1 .

If the former holds, then $F \leq E_0$ implies $F \geq \eta$ for all $\eta < \xi_0$, and so $F \geq \xi_0$, that is, $E_0 \underset{p}{\geq} \xi_0$. Similarly $E - E_0 \underset{p}{\leq} \xi_0$. So in this case the rest of (iv) is proved, too.

If the latter holds, then we have even a $\xi_1 - \delta$ in Γ' and $\xi_1 + \delta$ in Γ'' for some suitable $\delta > 0$. So an $F_1 \leq E_0$ with $F_1 < \xi_1 - \delta$ exists, and an $F_2 \leq E - E_0$ with

$F_2 > \xi_1 + \delta$. By Lemma 1.2 there exist two F_3 and F_4 (both $\neq 0$), with $F_3 \leq F_1$, $F_4 \leq F_2$ and $F_3 \leq \xi_1 - \delta$, $F_4 \geq \xi_1 + \delta$. We may assume that $D_M(F_3) = D_M(F_4)$, since otherwise we may replace F_3, F_4 by two $F'_3 \leq F_3, F'_4 \leq F_4$ with $D_M(F'_3) = D_M(F'_4) = \min(D_M(F_3), D_M(F_4))$ (remembering Lemma 1.2.3).

Now as $F_3 \leq E_0, F_4 \leq E - E_0$, so $E_0 - F_3 + F_4$ is a projection $\leq E$, and

$$D_M(E_0 - F_3 + F_4) = D_M(E_0) - D_M(F_3) + D_M(F_4) = D_M(E_0) = \lambda,$$

while

$$\begin{aligned} ((E_0 - F_3 + F_4)g, g) &= (E_0g, g) - (F_3g, g) + (F_4g, g) \\ &\geq m_a - (\xi_1 - \delta)D_M(F_3) + (\xi_1 + \delta)D_M(F_4) \\ &\geq m_a - (\xi_1 - \delta)D_M(F_3) + (\xi_1 + \delta)D_M(F_3) \\ &= m_a + 2\delta D_M(F_3) > m_a, \end{aligned}$$

contradicting the maximum property of m_a . So this case cannot arise.

The proof is therefore completed.

Lemma 3.4.2. *We can define for all $0 \leq \alpha \leq 1$ a family of projections $E(\alpha)$ and a function $\xi(\alpha)$ with the following properties:*

- (i) $E(0) = 0, \quad E(1) = 1,$
- (ii) $\alpha \leq \beta$ implies $E(\alpha) \leq E(\beta),$
- (iii) $\xi(0) = K, \quad \xi(1) = K^{-1},$
- (iv) $\alpha \leq \beta$ implies $\xi(\alpha) \geq \xi(\beta),$
- (v) $D_M(E(\alpha)) = \alpha,$
- (vi) $E(\alpha)$ reduces $g,$
- (vii) $\alpha < \beta$ implies $\xi(\beta - 0) \leq E(\beta) - E(\alpha) \leq \xi(\alpha + 0).$

Choose a sequence $\rho_1, \rho_2, \dots$ which lies and is everywhere dense in $0 \leq \alpha \leq 1$, with $\rho_1 = 0, \rho_2 = 1$. We will define $E(\alpha)$ and $\xi(\alpha)$ for $\alpha = \rho_1, \rho_2, \dots$ so that they fulfill (i)–(vi) and

$$(vii)' \quad E(\alpha) \underset{p}{\geq} \xi(\alpha), \quad 1 - E(\alpha) \underset{p}{\leq} \xi(\alpha),$$

for all $\alpha = \rho_1, \rho_2, \dots$.

Put first $E(0) = 0, E(1) = 1, \xi(0) = K, \xi(1) = K^{-1}$. Then $\alpha = \rho_1, \rho_2$ are taken care of, and (i)–(vi) as well as (vii)' hold for ρ_1, ρ_2 . Assume now that for a $j = 3, 4, \dots$ the $E(\alpha), \xi(\alpha)$ for $\alpha = \rho_1, \dots, \rho_{j-1}$ are already defined, so that (i)–(vi), (vii)' hold for $\alpha = \rho_1, \dots, \rho_{j-1}$; we will now define $E(\rho_j), \xi(\rho_j)$ without violating (i)–(vi), (vii)'.

Consider the $\rho_i, i = 1, \dots, j-1$ with $\rho_i \leq \rho_j$ (they exist: $\rho_1 = 0 \leq \rho_j$); let $\rho_{i'}$ be the greatest one. Consider the $\rho_i, i = 1, \dots, j-1$, with $\rho_i \geq \rho_j$ (they exist: $\rho_2 = 1 \geq$

ρ_j); let $\rho_{i''}$ be the smallest one. As $\rho_{i'}, \rho_{i''} \neq \rho_j$, so we have $\rho_{i'} < \rho_j < \rho_{i''}$. So (ii) gives $E(\rho_{i'}) \leq E(\rho_{i''})$, and (v) gives $D_M(E(\rho_{i''}) - E(\rho_{i'})) = \rho_{i''} - \rho_{i'} > \rho_j - \rho_{i'} > 0$.

Apply now [Lemma 3.4.1](#) to $E = E(\rho_{i''}) - E(\rho_{i'})$, $\lambda = \rho_j - \rho_{i'}$, with $a = \xi(\rho_{i''})$, $b = \xi(\rho_{i'})$. Owing to (vii)' and [Lemma 1.2.3](#) we have $E(\rho_{i''}) - E(\rho_{i'}) \leq E(\rho_{i''})$ and therefore $\geq \xi(\rho_{i''})$, and $E(\rho_{i''}) - E(\rho_{i'}) \leq 1 - E(\rho_{i'})$ and therefore $\leq \xi(\rho_{i'})$. That is, $a \leq_p E \leq_p b$. An $E_0 \leq E(\rho_{i''}) - E(\rho_{i'})$ and a ξ_0 result; put $E(\rho_j) = E(\rho_{i'}) + E_0$ (this is a projection in M) and $\xi(\rho_j) = \xi_0$. Let us now consider (i)–(vi), (vii)' for $\alpha = \rho_1, \dots, \rho_{j-1}, \rho_j$.

(i), (iii) are unaffected.

In (ii), (iv) the only new possibilities are $\alpha = \rho_i \leq \rho_j = \beta$ and $\alpha = \rho_j \leq \rho_i = \beta$, where $i = 1, \dots, j-1$. In the first case $\rho_i \leq \rho_{i'}$, so $E(\alpha) = E(\rho_i) \leq E(\rho_{i'}) \leq E(\rho_{i'}) + E_0 = E(\rho_j) = E(\beta)$, and $\xi(\alpha) = \xi(\rho_i) \geq \xi(\rho_{i'}) = b \geq \xi_0 = \xi(\rho_j) = \xi(\beta)$. In the second case $\rho_i \geq \rho_{i''}$, so $E(\beta) = E(\rho_i) \geq E(\rho_{i''}) = E(\rho_{i'}) + E(\rho_{i''}) - E(\rho_{i'}) \geq E(\rho_{i'}) + E_0 = E(\rho_j) = E(\alpha)$, and $\xi(\beta) = \xi(\rho_i) \leq \xi(\rho_{i''}) = a \leq \xi_0 = \xi(\rho_j) = \xi(\alpha)$. So (ii), (iv) remain true.

In (v) we need only $D_M(E(\rho_j)) = \rho_j$. Now $D_M(E(\rho_j)) = D_M(E(\rho_{i'}) + E_0) = D_M(E(\rho_{i'})) + D_M(E_0) = \rho_{i'} + (\rho_j - \rho_{i'}) = \rho_j$.

In (vi) we need only that $E(\rho_j)$ reduces g . As $E(\rho_j) = E(\rho_{i'}) + E_0$, and $E(\rho_{i'})$ reduces g , we must only show that E_0 reduces g (use [Lemma 3.3.3](#)). But E_0 reduces g with respect to $E = E(\rho_{i''}) - E(\rho_{i'})$ by [Lemma 3.4.1](#), (iii), and $E(\rho_{i''}) - E(\rho_{i'})$ reduces g because $E(\rho_{i''})$ and $E(\rho_{i'})$ do (again use [Lemma 3.3.3](#)), therefore E_0 reduces g by [Lemma 3.3.2](#). So the property (vi) remains true.

In (vii)' we need only $E(\rho_j) \geq_p \xi(\rho_j)$, $1 - E(\rho_j) \leq_p \xi(\rho_j)$. Now $E(\rho_j) = E(\rho_{i'}) + E_0$, and $E(\rho_{i'}) \geq_p \xi(\rho_{i'}) \geq \xi(\rho_j)$, $E_0 \geq_p \xi_0 = \xi(\rho_j)$ by [Lemma 3.4.1](#), (iv), so $E(\rho_j) \geq_p \xi(\rho_j)$ by [Lemma 3.3.6](#). On the other hand $1 - E(\rho_j) = 1 - E(\rho_{i''}) + ((E(\rho_{i''}) - E(\rho_{i'})) - E_0)$ and by [Lemma 3.4.1](#), (iv), $1 - E(\rho_{i''}) \leq_p \xi(\rho_{i''}) \leq \xi(\rho_j)$, $(E(\rho_{i''}) - E(\rho_{i'})) - E_0 = E - E_0 \leq_p \xi_0 = \xi(\rho_j)$, so that $1 - E(\rho_j) \leq_p \xi(\rho_j)$ by [Lemma 3.3.6](#). Therefore (vii)' remains true.

Thus we have verified (i)–(vi), (vii)' for $\alpha = \rho_1, \dots, \rho_{j-1}, \rho_j$. Therefore all $\alpha = \rho_1, \rho_2, \dots$ are taken care of, and (i)–(vi), (vii)' hold for all of them. Owing to (ii) and (iv) $\lim_{\rho_i \rightarrow \alpha} E(\rho_i)$ and $\lim_{\rho_i \rightarrow \alpha} \xi(\rho_i)$ exist for all α with $0 \leq \alpha \leq 1$. If α is equal to a ρ_j , then this limit is meant to denote the value at ρ_j . So we can extend the

definitions of $E(\alpha)$ and $\xi(\alpha)$ to all above α by defining

$$E(\alpha) = \lim_{\substack{\rho_i \rightarrow \alpha \\ \rho_i \geq \alpha}} E(\rho_i), \quad \xi(\alpha) = \lim_{\substack{\rho_i \rightarrow \alpha \\ \rho_i \geq \alpha}} \xi(\rho_i).$$

The statements (i), (iii) are unaffected by this extension, while (ii), (iv)–(vi) extend by continuity to all $0 \leq \alpha \leq 1$.

Let us now consider (vii). Assume $\alpha < \beta$. Choose a sequence ρ_{i_n} , $n = 0, \pm 1, \pm 2, \dots$, with $\alpha < \dots < \rho_{i_{-2}} < \rho_{i_{-1}} < \rho_{i_0} < \rho_{i_1} < \rho_{i_2} < \dots < \beta$ and $\lim_{n \rightarrow -\infty} \rho_{i_n} = \alpha$, $\lim_{n \rightarrow \infty} \rho_{i_n} = \beta$. We have $\rho_{i_{-1}} > \rho_{i_{-2}} > \dots > \alpha$, so that $E(\rho_{i_{-1}}) \geq E(\rho_{i_{-2}}) \geq \dots \geq E(\alpha)$. Therefore $\lim_{n \rightarrow -\infty} E(\rho_{i_n})$ exists, and is $\geq E(\alpha)$. Now we have

$$\begin{aligned} D_M \left(\lim_{n \rightarrow -\infty} E(\rho_{i_n}) - E(\alpha) \right) &= \lim_{n \rightarrow -\infty} D_M(E(\rho_{i_n})) - D_M(E(\alpha)) \\ &= \lim_{n \rightarrow -\infty} \rho_{i_n} - \alpha = \alpha - \alpha = 0, \\ \lim_{n \rightarrow -\infty} E(\rho_{i_n}) &= E(\alpha). \end{aligned}$$

Similarly

$$\lim_{n \rightarrow \infty} E(\rho_{i_n}) = E(\beta).$$

Therefore

$$\begin{aligned} \sum_{m=-\infty}^{\infty} (E(\rho_{i_{m+1}}) - E(\rho_{i_m})) &= \lim_{n \rightarrow \infty} \sum_{m=-n}^{n-1} (E(\rho_{i_{m+1}}) - E(\rho_{i_m})) \\ &= \lim_{n \rightarrow \infty} (E(\rho_{i_n}) - E(\rho_{i_{-n}})) = E(\beta) - E(\alpha). \end{aligned}$$

We have $E(\rho_{i_{m+1}}) - E(\rho_{i_m}) \leq E(\rho_{i_{m+1}})$, therefore (vii)' (for $\alpha = \rho_{i_{m+1}}$) and [Lemma 1.2.3](#) give $E(\rho_{i_{m+1}}) - E(\rho_{i_m}) \geq_p \xi(\rho_{i_{m+1}}) \geq \xi(\beta - 0)$. Then [Lemma 3.3.6](#) gives $E(\beta) - E(\alpha) \geq_p \xi(\beta - 0)$. On the other hand we have $E(\rho_{i_{m+1}}) - E(\rho_{i_m}) \leq 1 - E(\rho_{i_m})$; therefore (vii)' (for $\alpha = \rho_{i_m}$) and [Lemma 1.2.3](#) give $E(\rho_{i_{m+1}}) - E(\rho_{i_m}) \leq_p \xi(\rho_{i_m}) \leq \xi(\alpha + 0)$. Then [Lemma 3.3.6](#) gives $E(\beta) - E(\alpha) \leq_p \xi(\alpha + 0)$.

Thus we have proved

$$\xi(\beta - 0) \leq_p E(\beta) - E(\alpha) \leq_p \xi(\alpha + 0),$$

that is, (vii).

We have established (i)–(vii) for all $0 \leq \alpha \leq 1$, and so the proof is completed.

Lemma 3.4.3. Let α_i , $i = 0, 1, \dots, n$ be a set of numbers such that $0 = \alpha_0 \leq \alpha_1 \leq \dots \leq \alpha_{n-1} \leq \alpha_n = 1$ and let λ_i , $i = 1, \dots, n$ be a set of positive real numbers. Put (with

the $E(\alpha)$, $\xi(\alpha)$ of [Lemma 3.4.2](#))

$$g' = \sum_{i=1}^n \lambda_i (E(\alpha_i) - E(\alpha_{i-1}))g,$$

$$C_1 = \max_{i=1, \dots, n} \lambda_i^2 \xi(\alpha_{i-1} + 0), \quad C_2 = \min_{i=1, \dots, n} \lambda_i^2 \xi(\alpha_i - 0).$$

Then for all definite $A \in M$

$$C_1 \text{Tr}_M(A) \geq (Ag', g') \geq C_2 \text{Tr}_M(A).$$

We have

$$\begin{aligned} (Ag', g') &= \left(A \sum_{i=1}^n \lambda_i (E(\alpha_i) - E(\alpha_{i-1}))g, \sum_{j=1}^n \lambda_j (E(\alpha_j) - E(\alpha_{j-1}))g \right) \\ &= \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j (A(E(\alpha_i) - E(\alpha_{i-1}))g, (E(\alpha_j) - E(\alpha_{j-1}))g) \\ &= \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j ((E(\alpha_j) - E(\alpha_{j-1}))A(E(\alpha_i) - E(\alpha_{i-1}))g, g). \end{aligned}$$

As every $E(\alpha_k) - E(\alpha_{k-1})$ reduces g (by [Lemma 3.4.2](#), (vi), and [Lemma 3.3.3](#)) and as $(E(\alpha_j) - E(\alpha_{j-1})) \cdot (E(\alpha_i) - E(\alpha_{i-1})) = 0$ for $j \neq i$, [Lemma 3.3.4](#) permits us to drop all terms with $i \neq j$ from the above sum $\sum_{i,j=1}^n$. Therefore it becomes $\sum_{i=1}^n \lambda_i^2 ((E(\alpha_i) - E(\alpha_{i-1}))A(E(\alpha_i) - E(\alpha_{i-1}))g, g)$.

Now $E(\alpha_i) - E(\alpha_{i-1}) \leq_p \xi(\alpha_{i-1} + 0)$ by [Lemma 3.4.2](#), (vii). Therefore it is evident from [Lemma 1.4.1](#) that $((E(\alpha_i) - E(\alpha_{i-1}))A(E(\alpha_i) - E(\alpha_{i-1}))g, g) \leq \xi(\alpha_{i-1} + 0) \text{Tr}_M((E(\alpha_i) - E(\alpha_{i-1}))A(E(\alpha_i) - E(\alpha_{i-1})))$. So

$$\begin{aligned} (Ag', g') &\leq \sum_{i=1}^n \lambda_i^2 \xi(\alpha_{i-1} + 0) \text{Tr}_M((E(\alpha_i) - E(\alpha_{i-1}))A(E(\alpha_i) - E(\alpha_{i-1}))) \\ &\leq C_1 \sum_{i=1}^n \text{Tr}_M((E(\alpha_i) - E(\alpha_{i-1}))A(E(\alpha_i) - E(\alpha_{i-1}))) \\ &\leq C_1 \text{Tr}_M(A) \end{aligned}$$

(use [Lemma 3.3.5](#)). Similarly $E(\alpha_i) - E(\alpha_{i-1}) \geq_p \xi(\alpha_i - 0)$ by [Lemma 3.4.2](#), (vii). Therefore [Lemma 1.4.1](#) gives $((E(\alpha_i) - E(\alpha_{i-1}))A(E(\alpha_i) - E(\alpha_{i-1}))g, g) \geq \xi(\alpha_i -$

0) $Tr_M((E(\alpha_i) - E(\alpha_{i-1}))A(E(\alpha_i) - E(\alpha_{i-1})))$. So

$$\begin{aligned} (Ag', g') &\geq \sum_{i=1}^n \lambda_i^2 \xi(\alpha_i - 0) Tr_M((E(\alpha_i) - E(\alpha_{i-1}))A(E(\alpha_i) - E(\alpha_{i-1}))) \\ &\geq C_2 \sum_{i=1}^n Tr_M((E(\alpha_i) - E(\alpha_{i-1}))A(E(\alpha_i) - E(\alpha_{i-1}))) = C_2 Tr_M(A) \end{aligned}$$

(use [Lemma 3.3.5](#)). Thus the proof is completed.

3.5. We can now take the decisive step.

Lemma 3.5.1. *Let g and K be as in [Theorem I](#). Let K' be any number such that $1 < K' \leq K$. Then there is a g' which is related to K' as g is to K in [Theorem I](#), and such that*

$$\|g' - g\| \leq (K - 1)K^{1/2}.$$

Let n be the smallest positive integer with $(K')^n \geq K$. Put $\theta = K^{1/n}$, so $1 < \theta \leq K'$. Choose the α_i , $i = 0, 1, \dots, n$ with $0 = \alpha_0 \leq \alpha_1 \leq \dots \leq \alpha_{n-1} \leq \alpha_n = 1$, so that $\xi(\alpha_i + 0) \leq \theta^{n-2i} \leq \xi(\alpha_i - 0)$. (For $i = 0$, $\theta^{n-2i} = \theta^n = K$; for $i = n$, $\theta^{n-2i} = \theta^{-n} = K^{-1}$.) Put $\lambda_i = \theta^{i-(n+1)/2}$. Now apply [Lemma 3.4.3](#). We have $\lambda_i^2 \xi(\alpha_{i-1} + 0) \leq \theta^{2i-(n+1)} \theta^{n-2(i-1)} = \theta \leq K'$ and $\lambda_i^2 \xi(\alpha_i - 0) \geq \theta^{2i-(n+1)} \cdot \theta^{n-2i} \geq \theta^{-1} \geq (K')^{-1}$, so $C_1 \leq K'$ and $C_2 \geq (K')^{-1}$. Therefore the g' of that lemma gives for every definite $A \in M$: $K' Tr_M(A) \geq (Ag', g') \geq (K')^{-1} Tr_M(A)$; that is, $K'(Ag', g') \geq Tr_M(A) \geq (K')^{-1}(Ag', g')$. So it is related to K' as g and K are in [Theorem I](#).

Compute now $\|g' - g\|$. The projections $E(\alpha_i) - E(\alpha_{i-1})$, $i = 1, \dots, n$, are mutually orthogonal, therefore the $(E(\alpha_i) - E(\alpha_{i-1}))g$, $i = 1, \dots, n$ are mutually orthogonal too. Now we have

$$\begin{aligned} \|g' - g\|^2 &= \left\| \sum_{i=1}^n \lambda_i (E(\alpha_i) - E(\alpha_{i-1}))g - \sum_{i=1}^n (E(\alpha_i) - E(\alpha_{i-1}))g \right\|^2 \\ &= \left\| \sum_{i=1}^n (\lambda_i - 1) (E(\alpha_i) - E(\alpha_{i-1}))g \right\|^2 \\ &= \sum_{i=1}^n (\lambda_i - 1)^2 \|(E(\alpha_i) - E(\alpha_{i-1}))g\|^2 \\ &\leq \max_{i=1, \dots, n} (\lambda_i - 1)^2 \sum_{i=1}^n \|(E(\alpha_i) - E(\alpha_{i-1}))g\|^2 \\ &= \max_{i=1, \dots, n} (\lambda_i - 1)^2 \|g\|^2. \end{aligned}$$

Now clearly

$$\max_{i=1, \dots, n} (\lambda_i - 1)^2 = (\lambda_n - 1)^2 = (\theta^{(n+1)/2} - 1)^2 \leq (\theta^n - 1)^2 = (K - 1)^2,$$

so that $\|g' - g\| \leq (K - 1)\|g\|$.

But $A = 1$ in [Theorem I](#) gives

$$\|g\|^2 = (g, g) \leq K \text{Tr}_M(1) = K, \quad \|g\| \leq K^{1/2}.$$

Therefore $\|g' - g\| \leq (K - 1)K^{1/2}$. Thus the proof is completed.

Let $K_i = 1 + 2^{-(i+2)}$, $i = 0, 1, 2, \dots$. We define inductively a sequence of elements g_i , $i = 0, 1, 2, \dots$. Let g_0 be related to K_0 as g is to k in [Theorem I](#). Suppose g_i has been defined and is related to K_i as in [Theorem I](#). Then in [Lemma 3.5.1](#) let $g = g_i$, $K = K_i$, $K' = K_{i+1}$. We then define g_{i+1} as g' . Then g_{i+1} and K_{i+1} are related as g and K in [Theorem I](#) and we also have

$$\|g_{i+1} - g_i\| \leq (K_i - 1)K_i^{1/2} = \frac{1}{2^{i+2}}K_i^{1/2} \leq \frac{1}{2^{i+1}}.$$

Then for $p > 0$

$$\begin{aligned} \|g_{n+p} - g_n\| &= \left\| \sum_{i=n}^{n+p-1} (g_{i+1} - g_i) \right\| \leq \sum_{i=n}^{n+p-1} \|g_{i+1} - g_i\| \\ &\leq \sum_{i=n}^{n+p-1} \frac{1}{2^{i+1}} \leq \sum_{i=n}^{\infty} \frac{1}{2^{i+1}} = \frac{1}{2^n}. \end{aligned}$$

This implies that the g_i 's satisfy the Cauchy condition. Let g be their limit, and let A again be definite and in M . Then, since A is bounded,

$$(Ag, g) = \lim_{i \rightarrow \infty} K_i(Ag_i, g_i) \geq \text{Tr}_M(A) \geq \lim_{i \rightarrow \infty} K_i^{-1}(Ag_i, g_i) = (Ag, g).$$

Thus, if $A \in M$ is definite, $(Ag, g) = \text{Tr}_M(A)$.

For $A \in M$, Hermitian, we have as in [§2.1](#), under Property II, $A = A_1 - A_2$, A_1 and A_2 , in M and positive definite. Property I of [§2.1](#) then yields $(Ag, g) = \text{Tr}_M(A)$. If A is merely in M , then [Definition 2.2.1](#) now implies that $(Ag, g) = \text{Tr}_M(A)$. We have thus demonstrated

Theorem II. *Let M be a factor in case II, with $\alpha \geq 1$. (In the normalization of [§1.1](#). In the normalization of [R.O.](#), Theorem VIII this means M, M' are either in case $\text{II}_1, \text{II}_\infty$ or in case II_1, II_1 with $C \leq 1$, standard normalization.) Then there exists a $g \in \mathfrak{H}$ such that*

$$(Ag, g) = \text{Tr}_M(A).$$

We now drop the restriction that $\alpha \geq 1$. Let M, M' be in case II_1, II_1 , $\alpha < 1$, and let m be an integer such that $m\alpha \geq 1$ or in the standard normalization of [R.O.](#), Theorem X, $C/m \leq 1$. Let $E_1, E_2, \dots, E_m$ be m projections each of which is in M , $D_M(E_i) = 1/m$, $E_i E_j = 0$ if $i \neq j$. Let the range of E_i be $\mathfrak{M}_i$. Let us recall [R.O.](#), §11.3 using $\mathfrak{M}_i$ for $\mathfrak{M}$.

Consider the $M_{(\mathfrak{M}_i)} = M_i$ and $M'_{(\mathfrak{M}_i)} = M'_i$ of Definition 11.3.1. By Lemma 11.3.2, these constitute a factorization in $\mathfrak{M}_i$. Now we can take for $A \in M$ and such that $E_i A = A E_i = A$, $D_{M_i}(A E_i) = D_M(A)$, and for $A' \in M'$, $D_{M'_i}(A E_i) = D_{M'}(A')$, D_{M_i} and $D_{M'_i}$ are dimension functions for M_i and M'_i respectively (Lemma 11.3.6).

The ranges of $D_{M_{(\mathfrak{M})}}$ and $D_{M'_{(\mathfrak{M})}}$ are by Lemma 11.3.7, the intervals $(0, 1/m)$, $(0, \alpha)$ respectively. So we can obtain the standard normalization of [R.O.](#), Theorem X by multiplying by m and $1/\alpha$ respectively. But in this latter normalization, by Lemma 11.4.3, $C = (D_M(E_i)/D_{M'}(1))C_0 = C_0/m \leq 1$, where D_M and $D_{M'}$ refer to the standard normalization for M and M' as in [R.O.](#), Theorem X.

By [Theorem II](#) above this implies that there is a $g_i^0 \in \mathfrak{M}_i$ such that, if $A_0 \in M_i$, then $\text{Tr}_{M_i}(A_0) = (A_0 g_i^0, g_i^0)$. But now if $A \in M$ is such that $E_i A = A E_i = A$, let $A_0 = A E_i$. From the above it will be seen that $\text{Tr}_M(A) = \frac{1}{m} \text{Tr}_{M_i}(A_0) = \frac{1}{m} (A_0 g_i^0, g_i^0) = \frac{1}{m} (A g_i^0, g_i^0) = (A g_i, g_i)$, if $g_i = (1/m)^{1/2} g_i^0$.

Now if $A \in M$, we have by Lemma 3.2.6

$$\begin{aligned} \text{Tr}_M(A) &= \sum_{i=1}^m \text{Tr}_M(E_i A E_i) = \sum_{i=1}^m (E_i A E_i g_i, g_i) \\ &= \sum_{i=1}^m (A E_i g_i, E_i g_i) = \sum_{i=1}^m (A g_i, g_i). \end{aligned}$$

If M, M' is in case $\text{I}_m, \text{I}_{m'}$, $m < \infty$, we may proceed as follows. By [R.O.](#), Lemma 8.6.1, $\mathfrak{H} = E_m \otimes \mathfrak{H}_2$, where E_m is an m -dimensional euclidean space. Let $\phi_1, \dots, \phi_m$ be a complete orthonormal set in E_m and $f \in \mathfrak{H}_2$ with $\|f\| = 1$. Then if we let $g_i = \phi_i \otimes f$, we easily verify that the above formula for Tr_M holds. Thus we have

Theorem III. *If M is a factor in a finite case, then there exists a finite number of elements $g_i \in \mathfrak{H}$ such that*

$$\text{Tr}_M(A) = \sum_{i=1}^m (A g_i, g_i).$$

Suppose that $A \in M$ is held fixed. Then consider the weak neighborhood $U = U(A; g_1, \dots, g_m; g_1, \dots, g_m; \epsilon/m)$. $X \in U$ implies $|(X - A)g_i, g_i| < \epsilon/m$ for $i =$

$1, \dots, m$. Hence if X is also in M , [Theorem III](#) implies $|Tr_M(X) - Tr_M(A)| < \epsilon$. This proves

Theorem IV. *If M is a factor in a finite case, $Tr_M(A)$ is weakly continuous.*

CHAPTER IV. THE ISOMORPHISM OF M, M' AND $\mathfrak{H}$ (FOR $\alpha = 1$)

4.1. We assume that M, M' is in case II_1, II_1 and $\alpha = 1$ or what is the same thing that $C = 1$ and we have the standard normalization. We know by the discussion of [R.O.](#), §§11.3 and 11.4 how all other cases II can be reduced to this case.

As $\alpha = 1$, [Theorem II](#) holds. We define

Definition 4.1.1. *A g which satisfies [Theorem II](#) is uniformly distributed with respect to M ; abbreviated g u.d.r. M .*

Lemma 4.1.1. *If g is u.d.r. M , then (i) $\|g\| = 1$, (ii) $E_g^{M'} = 1$, (iii) $E_g^M = 1$ (cf. [R.O.](#), Definition 5.1.1).*

To prove (i), in [Theorem II](#) take A as 1. Then $1 = Tr_M(1) = (g, g) = \|g\|^2$.

To prove (ii), in [Theorem II](#) take A as $E_g^{M'}$. Then

$$1 = \|g\|^2 = (E_g^{M'} g, g) = Tr_M(E_g^{M'}) = D_M(E_g^{M'})$$

or $D_M(E_g^{M'}) = 1$ which in our normalization implies $E_g^{M'} = 1$.

Consider (iii). Since we have the standard normalization and $C = 1$, $D_{M'}(E_g^M) = D_M(E_g^{M'}) = 1$. This implies $E_g^M = 1$.

Definition 4.1.2. *Let g be in $\mathfrak{H}$. Let $\mathcal{Q}_g(M)$ consist of all operators $Z \in U(M)$, i.e., Z is linear closed ηM and with a dense domain (cf. [R.O.](#), Definition 4.2.1), for which Zg exists.*

Now consider any $f \in \mathfrak{H}$ such that $E_f^{M'} = E_f^M = 1$. By [R.O.](#), Lemma 9.2.1, if $h \in \mathfrak{H}$, then $h = XYf$, where X and Y are $\in U(M)$. But if M is in case II_1 by [R.O.](#), Theorem XV, $Z = [XY]$ is $\in U(M)$, and, since $Z \supseteq XY$, Zf exists, and we also have $h = XYf = Zf$; i.e., every $h \in \mathfrak{H}$ is in the form Zf , where Z is $\in U(M)$.

Furthermore this Z is uniquely determined by h . For suppose Z_1 and $Z_2 \in U(M)$ are such that $Z_1f = Z_2f$ or $(Z_1 - Z_2)f = 0$. Then $[Z_1 - Z_2]$ is by [R.O.](#), Theorem XV, $\in U(M)$ and since $[Z_1 - Z_2] \supseteq Z_1 - Z_2$, $[Z_1 - Z_2]f$ exists and is zero. Let A' be $\in M'$. Then since $A'[Z_1 - Z_2] \subseteq [Z_1 - Z_2]A'$, $[Z_1 - Z_2]A'f = A'[Z_1 - Z_2]f = 0$. Thus $[Z_1 - Z_2]$ is zero on the set of $A'f$, $A' \in M$. But since $E_f^{M'} = 1$, this latter set is everywhere dense in $\mathfrak{H}$. Thus $[Z_1 - Z_2]$ is zero on a dense set and since it is closed it must be zero. [R.O.](#), Theorem XV, then yields

$$Z_1 = [Z_1 - 0] = [Z_1 - [Z_1 - Z_2]] = [[Z_1 - Z_1] + Z_2] = [0 + Z_2] = Z_2.$$

So we have proved

Lemma 4.1.2. *If f is such that $E_f^{M'} = E_f^M = 1$ (in particular, if f is u.d.r. M), then $h = Zf$ defines a one-to-one correspondence of $\mathfrak{H}$ and the set of all $Z \in U(M)$ for which Zf exists (i.e., between $\mathfrak{H}$ and $\mathcal{Q}_f(M)$).*

Since our assumptions on f are symmetric in M and M' , we also have

Lemma 4.1.3. *Let f be as in Lemma 4.1.2. Then $h = Z'f$ defines a one-to-one correspondence of $\mathfrak{H}$ and the set of all $Z' \in U(M')$ for which $Z'f$ exists (i.e., between $\mathfrak{H}$ and $\mathcal{Q}_f(M')$).*

Thus for $h \in \mathfrak{H}$ we have three correspondences defined by the equations $Zf = h = Z'f$, if f is u.d.r. M : (1) the correspondence $\mathfrak{I}_M$ of $\mathfrak{H}$ and $\mathcal{Q}_f(M)$, (2) the correspondence $\mathfrak{I}_{M'}$ of $\mathfrak{H}$ and $\mathcal{Q}_f(M')$, and (3) the correspondence $\mathfrak{I}_{M,M'}$ of $\mathcal{Q}_f(M)$ and $\mathcal{Q}_f(M')$.

4.2. We now investigate these three correspondences. We know that they are one-to-one. They are obviously linear and of course along with $\mathfrak{H}$ the sets $\mathcal{Q}_f(M)$ and $\mathcal{Q}_f(M')$ are linear. But inasmuch as $\mathfrak{I}_{M,M'}$ is a correspondence of operators, we would like to know the algebraic properties of this correspondence as well as certain properties of $\mathcal{Q}_f(M)$ and $\mathcal{Q}_f(M')$. In particular we would like to know:

- (i) To what extent can the operation $[XY]$ be performed in $\mathcal{Q}_f(M)$ (or $\mathcal{Q}_f(M')$)?
- (ii) To what extent can the operation X^* be performed in $\mathcal{Q}_f(M)$ (or $\mathcal{Q}_f(M')$)?
- (iii) Is the property of being bounded invariant under $\mathfrak{I}_{M,M'}$; i.e., since $M \subset \mathcal{Q}_f(M)$, $M' \subset \mathcal{Q}_f(M')$, is M mapped on M' by $\mathfrak{I}_{M,M'}$?
- (iv) Are the properties of being Hermitian, definite, or unitary invariant under $\mathfrak{I}_{M,M'}$?

Lemma 4.2.1. *If f is u.d.r. M , then every Uf , $U \in M$ and unitary, is u.d.r. M also. If $A \in M$, then*

$$(AUf, Uf) = (U^{-1}AUf, f) = \text{Tr}_M(U^{-1}AU) = \text{Tr}_M(A).$$

Lemma 4.2.2. *If f and g are each u.d.r. M , then there exists a $U' \in M'$ and unitary with $g = U'f$.*

If $h = Af$, $A \in M$, define h^* as Ag . The correspondence $h \rightleftharpoons h^*$ is linear, and, since

$$\begin{aligned} \|h^*\|^2 &= \|Ag\|^2 = (Ag, Ag) = (A^*Ag, g) = \text{Tr}_M(A^*A) = (A^*Af, f) \\ &= (Af, Af) = \|Af\|^2 = \|h\|^2, \end{aligned}$$

it is isometric. Thus the equation $Wh = h^*$ defines a linear isometric and hence

one-valued operator W . Since $E_f^M = E_g^M = 1$, both domain and range of W are everywhere dense and hence its closure $\widetilde{W}$ is unitary.

Now if h is in the domain of W , $h = Af$, $A \in M$. Then, for every $B \in M$, $BWh = Bh^* = BAf = WBf = WBh$ or $BW \subseteq WB$. This implies that $[BW] \subseteq [WB]$. But $B\widetilde{W} = \widetilde{BW} \subseteq [BW]$, while since B is bounded $[WB] = \widetilde{WB}$. Hence $B\widetilde{W} \subseteq \widetilde{WB}$. If in particular B is unitary, **R.O.**, Lemma 4.2.2 now yields that $\widetilde{W}$ is $\eta M'$ and **R.O.**, Lemma 4.2.1 implies that $\widetilde{W} \in M'$.

Letting $A = 1$ in the definition of W , we get $g = Wf = \widetilde{W}f$.

Lemma 4.2.3. *If f and g are u.d.r. M , then there exists a $U \in M$ and unitary such that $g = Uf$.*

By **Lemma 4.1.2**, $g = Zf$ where $Z \in U(M)$. Using the canonical decomposition for Z , $Z = BW$, where B is self-adjoint and definite and W is partially isometric and as indicated in the proof of **Lemma 1.4.3** may be taken as unitary; that is, $g = BWf$, where W is unitary and B definite and self-adjoint. We must show that $B = 1$.

By **Lemma 4.2.1**, $f_0 = Wf$ is u.d.r. M . Hence we must show that if f_0 and g are u.d.r. M and $g = Bf_0$, where B is definite and self-adjoint, then $B = 1$. Let $E(\lambda)$ be the resolution of the identity corresponding to B . If $E(\lambda) = 0$ for $\lambda < 1$ and $E(\lambda) = 1$ for $\lambda > 1$, then $B = 1$. Thus if B is not 1, either there exists a $\lambda_1 < 1$ for which $E(\lambda_1) \neq 0$ or a $\lambda_2 > 1$ for which $E(\lambda_2) \neq 1$. In the first case we have $\|E(\lambda_1)f_0\|^2 = (E(\lambda_1)f_0, E(\lambda_1)f_0) = (E(\lambda_1)f_0, f_0) = \text{Tr}_M(E(\lambda_1)) = (E(\lambda_1)g, g) = \|E(\lambda_1)g\|^2 = \|E(\lambda_1)Bf_0\|^2 = \|BE(\lambda_1)f_0\|^2 \leq \lambda_1^2\|E(\lambda_1)f_0\|^2$ or $(1 - \lambda_1^2)\|E(\lambda_1)f_0\|^2 \leq 0$. This implies $0 = \|E(\lambda_1)f_0\|^2 = (E(\lambda_1)f_0, f_0) = \text{Tr}_M(E(\lambda_1)) = D_M(E(\lambda_1))$ or $E(\lambda_1) = 0$, a contradiction. In the second case a similar type of calculation yields that $\|(1 - E(\lambda_2))f_0\|^2 = \|B(1 - E(\lambda_2))f_0\|^2 \geq \lambda_2^2\|(1 - E(\lambda_2))f_0\|^2$, which again implies that $1 - E(\lambda_2) = 0$, a contradiction. Thus B must be 1 and our lemma is proved.

Lemma 4.2.4. *If f is u.d.r. M , and $U' \in M'$ and unitary, then $U'f$ is u.d.r. M .*

If $A \in M$,

$$(AU'f, U'f) = (U'^{-1}AU'f, f) = (AU'^{-1}U'f, f) = (Af, f) = \text{Tr}_M(A).$$

Lemma 4.2.5. *Unitary operators correspond to unitary operators under $\mathfrak{S}_{M,M'}$; i.e., if f is u.d.r. M , $U \in M$, $U' \in M'$ and $Uf = U'f$, then if either U or U' is unitary the other is too.*

Let $U \in M$ be unitary, then by **Lemma 4.2.1** Uf is u.d.r. M . Then **Lemma 4.2.2** yields that there is a unitary $U' \in M$, such that $U'f = Uf$. Thus $U \sim U'$ under $\mathfrak{S}_{M,M'}$. Similarly **Lemmas 4.2.4** and **4.2.3** yield that to every unitary $U' \in M$, there exists a unitary $U \in M$ such that $U'f = Uf$.

Lemma 4.2.6. f u.d.r. M is equivalent to f u.d.r. M' .

Let f be u.d.r. M and $A' \in M$. Then consider $T_0(A') = (A'f, f)$. It is linear in A' . By [Lemma 4.1.1](#), (i) $T_0(1) = 1$. If A' is definite, then $T_0(A') = (A'f_0, f_0) \geq 0$. Furthermore we have $T_0(A'^*) = (A'^*f, f) = (f, A'f) = \overline{(A'f, f)} = \overline{T_0(A)}$. Therefore $T_0(A')$ satisfies (i)–(iv) and (v) of [Property III](#) with M' instead of M .

We would like to verify next (vi)' of the same property for $T_0(A')$. By [Lemma 4.2.5](#) we have for any unitary $U' \in M$

$$\begin{aligned} T_0(U'^{-1}A'U') &= (U'^{-1}A'U'f, f) = (A'U'f, U'f) = (A'Uf, Uf) \\ &= (U^{-1}A'Uf, f) = (A'(U^{-1}U)f, f) = (A'f, f) = T_0(A'). \end{aligned}$$

Thus (vi)' holds. [Property IV](#) of §2.2 now implies that $T_0(A') = \text{Tr}_{M'}(A')$ and hence f is u.d.r. M' .

Replacing M by M' in the above argument yields the converse.

We note that the above argument also proves

Lemma 4.2.7. If $f \in \mathfrak{H}$ is such that (i) $\|f\| = 1$ and (ii) if to every unitary $U \in M$ there exists a unitary $U' \in M'$ such that $Uf = U'f$, then f is u.d.r. M .

Theorem V. Let $f \in \mathfrak{H}$ be such that $\|f\| = 1$ and let M, M' be in case II_1, II_1 with $C = 1$. Then the following statements are equivalent:

- (α) f is u.d.r. M .
 - (β) f is u.d.r. M' .
 - (γ) If U is unitary and $\in M$, then there exists a unitary $U' \in M'$ such that $Uf = U'f$.
 - (δ) If U' is unitary and $\in M'$, then there exists a unitary $U \in M$ such that $U'f = Uf$.
- Furthermore either (α), (β), (γ), or (δ) implies $E_f^M = E_f^{M'} = 1$.

(α) is equivalent to (β) by [Lemma 4.2.6](#). (α) and (γ) are equivalent by [Lemmas 4.2.5](#) and [4.2.7](#). Replacing M by M' , the last mentioned lemmas also show (β) and (δ) to be equivalent. This and [Lemma 4.1.1](#) prove the last statement of the theorem.

In view of [Theorem V](#), we will say that f is uniformly distributed (abbreviated u.d.) if f is u.d.r. M .

Corollary. If f is u.d., $\mathfrak{S}_{M, M'}$ maps $M(\subseteq \mathcal{Q}_f(M))$ on $M'(\subseteq \mathcal{Q}_f(M'))$; that is, the property of being bounded is invariant under $\mathfrak{S}_{M, M'}$.

By [Theorem V](#), the property of being unitary is invariant under $\mathfrak{S}_{M, M'}$. This and the linearity of $\mathfrak{S}_{M, M'}$ imply together that the property of having the form $A = \sum_{v=1}^n a_v U_v$ ($n = 1, 2, \dots$; a_v complex, U_v unitary and in M or M' along with A) is invariant under $\mathfrak{S}_{M, M'}$.

We will show that $A \in \mathcal{Q}_f(M)$ (or $\mathcal{Q}_f(M')$) is bounded if and only if it is in this form. Now if A is in this form, it is obviously bounded so we must only show that

every bounded A is in this form. Since $A = A_1 + iA_2$, A_1 and A_2 Hermitian, it will be sufficient if this is shown for Hermitian A 's. Since if A is in this form, cA is also, we may assume that the bound of A is ≤ 1 . Then $1 - A^2$ is definite and $(1 - A^2)^{1/2}$ exists. Letting $U = A + i(1 - A^2)^{1/2}$, then $U^* = A - i(1 - A^2)^{1/2}$ and U is unitary. Furthermore $A = \frac{1}{2}(U + U^*)$.

Now since the form $\sum_{v=1}^n a_v U_v$ is invariant under $\mathfrak{S}_{M,M'}$, boundedness must be also.

Theorem VI. Assume that f_0 is u.d. Then $\mathfrak{S}_{M,M'}$ is an anti-isomorphism of M and M' . That is, if $A \sim A'$, $B \sim B'$, then (i) $\alpha A \sim \alpha A'$; (ii) $A^* \sim A'^*$; (iii) $A + B \sim A' + B'$; (iv) $AB \sim B'A'$; (v) if A (or A') is Hermitian, then A' (or A) is also.

(i) and (iii) are obvious.

Consider (iv). We have $ABf_0 = AB'f_0 = B'Af_0 = B'A'f_0$, so that $AB \sim B'A'$. To prove (v), let U be unitary, $U \sim U'$. Then U' is unitary also by Theorem V. Similarly inasmuch as U^{-1} is unitary, if $U^{-1} \sim V'$, V' is unitary. As $UU^{-1} = U^{-1}U = 1$ by (iv) we have $U'V' = V'U' = 1$, and hence $V' = (U')^{-1}$. So $U^{-1} \sim (U')^{-1}$ and $\alpha(U + U^{-1}) \sim \alpha(U' + (U')^{-1})$ by (iii). If α is > 0 , the right side is Hermitian and the left side is any Hermitian element of M (cf. the proof of the corollary of Theorem V). This proves (v).

We now prove (ii). If A is $\in M$, then $A = B + iC$, $A^* = B - iC$, B, C are Hermitian and $\in M$. If $A \sim A'$, $B \sim B'$, $C \sim C'$, then B', C' are Hermitian and $A \sim B' + iC' = A'$, $A^* \sim B' - iC' = A'^*$, proving (ii).

Corollary. The properties of being Hermitian, being definite, being a projection, as well as the numerical quantities $\|A\|$ (the bound of A) and $Tr_M(A)$ (or $Tr_{M'}(A')$), all within M (or M'), are invariant under $\mathfrak{S}_{M,M'}$.

A Hermitian means $A = A^*$; A a projection means $A = A^* = A^2$; A definite means that there exists an Hermitian B with $A = B^2$. So all these properties are invariant under $\mathfrak{S}_{M,M'}$. 1 is invariant ($1 \in M$ and $1 \in M'$, $1f_0 = 1f_0$). $\|A\|$ is the smallest $\alpha \geq 0$ such that $\alpha^2 \cdot 1 - A^*A$ is definite, so $\|A\|$ is invariant. If $A \sim A'$, then $Af_0 = A'f_0$ and $Tr_M(A) = (Af_0, f_0) = (A'f_0, f_0) = Tr_M(A')$ by Theorem V.

4.3. In what follows f_0 will always be assumed to be u.d. The discussion which follows could be based on an extension of the notion of $Tr_M(A)$ (and of $Tr_{M'}(A')$) to unbounded $A \eta M$ (or $A' \eta M'$) but we prefer an approach which avoids this.

Theorem VII. The sets $\mathcal{Q}_{f_0}(M)$ and $\mathcal{Q}_{f_0}(M')$ are independent of the choice of u.d. f_0 . We will therefore denote them by $\mathcal{Q}(M)$ and $\mathcal{Q}(M')$, respectively. Furthermore the values of $\|Zf_0\|$, (Xf_0, Yf_0) , where $X, Y, Z \in \mathcal{Q}(M)$, or $\mathcal{Q}(M')$, are independent of the

choice of the u.d. f_0 .

Owing to the symmetry between M and M' it suffices to prove the statements concerning M ; i.e., let f_0 and g_0 be u.d., then for $A \in U(M)$, Af_0 is defined if and only if Ag_0 is defined and $\|Af_0\| = \|Ag_0\|$. Owing to the symmetry in f_0 and g_0 it will be sufficient to show that Ag_0 is defined if Af_0 is and $\|Af_0\| = \|Ag_0\|$.

By [Lemma 4.2.2](#), there exists a unitary $U' \in M'$ such that $g_0 = U'f_0$. Since $A \in U(M)$, $A = U'^{-1}AU'$. Hence since Af_0 exists $Af_0 = U'^{-1}AU'f_0 = U'^{-1}Ag_0$ exists. This implies that Ag_0 exists and since $Ag_0 = U'Af_0$, $\|Ag_0\| = \|Af_0\|$. Also if $B \in Q_{f_0}(M)$, then $Bg_0 = U'Bf_0$ and $(Ag_0, Bg_0) = (U'Af_0, U'Bf_0) = (Af_0, Bf_0)$.

So we must have these mappings:

$$(I) \quad \mathfrak{I}_M : \mathfrak{H} \sim Q(M); \quad \mathfrak{I}_{M'} : \mathfrak{H} \sim Q(M'); \quad \mathfrak{I}_{M,M'} : Q(M) \sim Q(M').$$

By the [corollary](#) to [Theorem V](#), $\mathfrak{I}_{M,M'}$ maps M on M' . So $\mathfrak{I}_M$ maps M and $\mathfrak{I}_{M'}$ maps M' on the same subset $\mathfrak{A}$ of $\mathfrak{H}$ and we have the further mappings

$$(II) \quad \mathfrak{I}_M : \mathfrak{A} \sim M; \quad \mathfrak{I}_{M'} : \mathfrak{A} \sim M'; \quad \mathfrak{I}_{M,M'} : M \sim M'.$$

(II) is part of (I).

We now prove

Lemma 4.3.1. $\mathfrak{A}$ is a linear set, dense in $\mathfrak{H}$.

Inasmuch as M is linear, $\mathfrak{A}$ is too.

Consider an $f \in \mathfrak{H}$, $f = Zf_0$, $Z \in Q(M)$. Now $Z = BU$, B self-adjoint and definite, $B \in U(M)$, U unitary and $U \in M$. Write B in its spectral form

$$B = \int_0^\infty \lambda dE(\lambda), \quad E(\lambda) \in M.$$

Then we have

$$[E(\mu)B] = \int_0^\mu \lambda dE(\lambda);$$

and therefore (i) $[E(\mu)B] \in M$ and (ii) if Bg is defined then we have strong $\lim_{\mu \rightarrow \infty} [E(\mu)B]g = Bg$. Thus $[E(\mu)B]U \in M$ and strong $\lim_{\mu \rightarrow \infty} [E(\mu)B]Uf_0 = BUf_0 = Zf_0 = f$. Since $[E(\mu)B]Uf_0 \in \mathfrak{A}$, f is a condensation point of $\mathfrak{A}$. As this is true for any $f \in \mathfrak{H}$, $\mathfrak{A}$ is dense in $\mathfrak{H}$.

Definition 4.3.1. If $f = Xf_0 = X'f_0$, $g = Yf_0 = Y'f_0$, X and $Y \in Q(M)$, X' and $Y' \in Q(M')$, let $\|X\| = \|X'\| = \|f\|$, $\langle\langle X, Y \rangle\rangle = \langle\langle X', Y' \rangle\rangle = (f, g)$.

[Theorem VII](#) shows that the values of $\|X\|$, $\|X'\|$, $\langle\langle X', Y' \rangle\rangle$, $\langle\langle X, Y \rangle\rangle$ are independent of the choice of the u.d. f_0 .

Lemma 4.3.2. If X and Y are $\in M$, then $\|X\| = (Tr_M(X^*X))^{1/2} = (Tr_M(XX^*))^{1/2}$, $\langle\langle X, Y \rangle\rangle = Tr_M(Y^*X) = Tr_M(XY^*)$.

As $\llbracket X \rrbracket = (\llbracket X, X \rrbracket)^{1/2}$, the second statement implies the first. Now clearly

$$\llbracket X, Y \rrbracket = (X f_0, Y f_0) = (Y^* X f_0, f_0) = \text{Tr}_M(Y^* X)$$

since f is u.d. We also have $\text{Tr}_M(Y^* X) = \text{Tr}_M(X Y^*)$ by §2.2, Property III, (vi). So $\llbracket X, Y \rrbracket = \text{Tr}_M(Y^* X) = \text{Tr}_M(X Y^*)$.

Lemma 4.3.3. *If $X', Y' \in M'$, then $\llbracket X' \rrbracket = (\text{Tr}_{M'}(X'^* X'))^{1/2} = (\text{Tr}_{M'}(X' X'^*))^{1/2}$, $\llbracket X', Y' \rrbracket = \text{Tr}_{M'}(Y'^* X') = \text{Tr}_{M'}(X' Y'^*)$.*

Replace M by M' in the proof of Lemma 4.3.2.

Theorem VIII. *If we use the definitions*

$$\begin{aligned} \llbracket X, Y \rrbracket &= \text{Tr}_M(Y^* X) = \text{Tr}_M(X Y^*), \\ \llbracket X \rrbracket &= (\llbracket X, X \rrbracket)^{1/2} = (\text{Tr}_M(X^* X))^{1/2} = (\text{Tr}_M(X X^*))^{1/2}, \end{aligned}$$

then M is an incomplete Hilbert space. Its completion in the usual (Cantor) way gives a Hilbert space $\tilde{M}$ which may be identified with $Q(M)$.

This is clear by the isomorphisms (I) and (II) and Lemmas 4.3.1 and 4.3.2. M and $\tilde{M}$, that is, $Q(M)$, are isomorphic to $\mathfrak{A}$ and $\mathfrak{S}$ respectively, using the isomorphism $\mathfrak{I}_M$.

The corresponding facts hold for M' ; their formulation is obvious.

Theorem IX. *$X \in Q(M)$ implies $X^* \in Q(M)$ and $\llbracket X \rrbracket = \llbracket X^* \rrbracket$.*

Assume $X \in Q(M)$, i.e., $X \in U(M)$; $X f_0$ is defined. Write $X = BU$, B self-adjoint and ηM , U unitary and $\in M$. Thus $BU f_0$ is defined. Now $U f_0$ is u.d. (Lemma 4.2.1), hence the existence of $BU f_0$ implies $B \in Q(M)$ (Theorem VII). This in turn implies the existence of $B f_0$ and with it $X^* f_0 = U^{-1} B f_0$. So $X^* \in Q(M)$.

As f_0 and $U f_0$ are both u.d. (Lemma 4.2.1) we have by Definition 4.3.1, $\llbracket X \rrbracket = \|X f_0\| = \|BU f_0\| = \|B\| = \|B f_0\| = \|U^{-1} B f_0\| = \|X^* f_0\| = \llbracket X^* \rrbracket$.

Corollary. *$X \sim X^*$ is an involutory conjugate anti-isomorphism of $Q(M)$ and isometric.*

$X \sim X^*$ maps $Q(M)$ on part of itself by Theorem IX. This and $X^{**} = X$ show that $X \sim X^*$ is a one-to-one and involutory mapping of $Q(M)$ on itself. It is isometric by Theorem IX; $\llbracket X \rrbracket = \llbracket X^* \rrbracket$ and it is obviously a conjugate anti-isomorphism.

The corresponding facts hold for M' ; the formulation is obvious.

4.4. We next discuss the algebra of $Q(M)$.

Property I⁰. *$A, B \in Q(M)$ imply $\alpha A, A^*, [A + B] \in Q(M)$. If also either A or B is $\in M$, then $[AB]$ is $\in Q(M)$.*

$\alpha A, [A + B]$ are $\in \mathcal{Q}(M)$ since $\mathcal{Q}(M)$ is linear (along with $\mathfrak{H}$). A^* is $\in \mathcal{Q}(M)$ by [Theorem IX](#).

Assume now that $A \in M, B \in \mathcal{Q}(M)$. Then Bf_0 is defined and so is $ABf_0 = [AB]f_0$. Thus $[AB] \in \mathcal{Q}(M)$. If $A \in \mathcal{Q}(M), B \in M$, then $B^* \in M, A^* \in \mathcal{Q}(M)$, so $[B^*A^*] \in \mathcal{Q}(M)$. Since $[B^*A^*]^* = [AB]$ ([R.O.](#), Theorem XV), $[AB]$ is $\in \mathcal{Q}(M)$.

Property II⁰. (i) $\|[\alpha A]\| = |\alpha| \cdot \|A\|$; (ii) $\|A^*\| = \|A\|$; (iii) $\|A + B\| \leq \|A\| + \|B\|$; (iv) $\|[AB]\| \leq \|A\| \cdot \|B\|$ and $\|A\| \cdot \|B\|$.

(i) and (iii) hold because $\mathcal{Q}(M)$ is isomorphic to $\mathfrak{H}$. (ii) holds by [Theorem IX](#).

Consider (iv). We have

$$\|[AB]\| = \|[AB]f_0\| = \|ABf_0\| \leq \|A\| \cdot \|Bf_0\| = \|A\| \cdot \|B\|$$

proving the first inequality. The second follows from this, by using (ii)

$$\|[AB]\| = \|[AB]^*\| = \|[B^*A^*]\| \leq \|B^*\| \cdot \|A^*\| = \|B\| \cdot \|A\|.$$

We have $\mathfrak{H} \sim \mathcal{Q}(M)$. Neither $\mathfrak{H}$ nor $\mathcal{Q}(M)$ depends on the choice of the u.d. f_0 but $\mathfrak{I}_M$ does. This dependence is as follows.

Property III⁰. Let us replace the u.d. f_0 by a u.d. g_0 and let $\widetilde{\mathfrak{I}}_M$ denote the resulting correspondence. Let U be unitary and $\in M$ and such that $Uf_0 = g_0$ (cf. [Lemma 4.2.1](#)). Then if $X = YU$, X and $Y \in \mathcal{Q}(M)$, we have

$$\mathfrak{I}_M : f \sim X; \quad \widetilde{\mathfrak{I}}_M : f \sim Y$$

if $f = Xf_0$.

This is clear since $f = Xf_0 = YUf_0 = Yg_0$.

We can now determine another notion which does not depend on the choice of the u.d. f_0 .

Property IV⁰. The linear set $\mathfrak{A}$ (cf. [§4.3](#), the correspondences (II)) does not depend on the choice of the u.d. f_0 .

This follows from the definition of $\mathfrak{A}$ and the fact that if $X = YU$, U unitary and either X or Y is bounded, then both X and Y are bounded.

Now $\mathcal{Q}(M) \sim \mathcal{Q}(M')$ under $\mathfrak{I}_{M,M'}$ and while neither $\mathcal{Q}(M)$ nor $\mathcal{Q}(M')$ depends on the choice of the u.d. f_0 yet $\mathfrak{I}_{M,M'}$ does. We now obtain this dependence.

Property V⁰. Let the u.d. f_0 be replaced by the u.d. g_0 and let the resulting correspondence between $\mathcal{Q}(M)$ and $\mathcal{Q}(M')$ be denoted by $\widetilde{\mathfrak{I}}_{M,M'}$. Then if $U \in M$ is unitary and such that $g_0 = Uf_0$, and $X = U^{-1}YU$, X and $Y \in \mathcal{Q}(M)$, then

$$\mathfrak{I}_{M,M'} : X' \sim X; \quad \widetilde{\mathfrak{I}}_{M,M'} : X' \sim Y$$

if $X'f_0 = Xf_0$, $X' \in Q(M')$.

Let $X'f_0 = Xf_0$, then $X'g_0 = X'Uf_0 = UX'f_0 = UXf_0 = UXU^{-1}g_0$.

Definition 4.4.1. The isomorphism $\mathfrak{S} \sim Q(M)$ makes correspond to every operator P in $\mathfrak{S}$ an operator P^0 in $Q(M)$.

Observe that inasmuch as the elements of $Q(M)$ are operators in $\mathfrak{S}$, P^0 is an operator on the operators of $\mathfrak{S}$.

Theorem X. (i) If $A \in M$, then

$$A^0Z = [AZ];$$

(ii) if $A' \in M'$ and $A' \sim A \in M$ under $\mathfrak{S}_{M,M'}$, then

$$A'^0 = [ZA].$$

P^0 is defined as follows. If $g = Xf_0$, $h = Pg$, $h = Yf_0$, X and $Y \in Q(M)$, then $P^0X = Y$. Thus to prove (i), we have $(A^0Z)f_0 = A(Zf_0) = [AZ]f_0$ and to show (ii) $(A'^0Z)f_0 = A'Zf_0 = ZA'f_0 = ZAf_0 = [ZA]f_0$.

Corollary 1. Let M^0 be the set of all operators L_A in $Q(M)$, $A \in M$, where $L_AZ = [AZ]$, and M_0 the set of all operators R_A in $Q(M)$, $A \in M$, where $R_AZ = [ZA]$. Then $\mathfrak{S}_M$ carries M into M^0 and M' into M_0 .

This is obvious by [Theorem X](#).

Corollary 2. M^0, M_0 are rings and $M^{0'} = M_0$ (all in $Q(M)$). They are factors of class (II_1, II_1) with $C = 1$.

Since $\mathfrak{S}_M$ is a spatial isomorphism of $\mathfrak{S}$ and $Q(M)$ which takes M, M' into M^0, M_0 respectively, these properties hold.

We have now characterized $\mathfrak{S}, M, M'$ by the situation in $Q(M), M^0, M_0$ to which they are spatially isomorphic. The spatial isomorphism is $\mathfrak{S}_M$ which depends on an arbitrary u.d. f_0 , while $Q(M), M^0, M_0$ themselves do not. All the influence of the choice of f_0 is however a further transformation R_U , $X = YU$ ($U \in M$, U unitary so $R_U \in M_0$). $\mathfrak{S}_M$ always carries the totality of u.d. elements g_0 of $\mathfrak{S}$ into the totality of unitary elements $U \in Q(M)$, f_0 being that element which corresponds to 1.

4.5. The following important isomorphism theorem can now be proved.

Theorem XI. Let $\mathfrak{S}_1$ and $\mathfrak{S}_2$ be two Hilbert spaces and M_1, M'_1 and M_2, M'_2 respectively be factor pairs in them, both of class (II_1, II_1) and with $C = 1$ in the standard normalization. Then an algebraic ring isomorphism of M_1 and M_2 is a necessary and sufficient condition for the existence of a spatial isomorphism of $\mathfrak{S}_1$ and $\mathfrak{S}_2$ which takes M_1, M'_1 into M_2, M'_2 .

The necessity is obvious and so we prove the sufficiency. Let $\mathfrak{F}$ be the algebraic ring isomorphism of M_1 and M_2 . By §2.2, Property IV, $Tr_{M_1}(A_1) = Tr_{M_2}(A_2)$ if $A_1 \sim A_2$ under $\mathfrak{F}$. So $\llbracket A_1 \rrbracket = \llbracket A_2 \rrbracket$, $\langle\langle A_1, B_1 \rangle\rangle = \langle\langle A_2, B_2 \rangle\rangle$, if $A_1 \sim A_2$, $B_1 \sim B_2$ under $\mathfrak{F}$. Thus $\mathfrak{F}$ is an isometric mapping of M_1 on M_2 and therefore extends by continuity in a unique way to a linear and isometric mapping of $\mathcal{Q}(M)$ on $\mathcal{Q}(M')$ which we call $\mathfrak{F}$ again. $A_1 \sim A_2$, $B_1 \sim B_2$ under $\mathfrak{F}$ imply $A_1 B_1 \sim A_2 B_2$ under $\mathfrak{F}$ if $A_1, B_1 \in M_1$ (and thus $A_2, B_2 \in M_2$). By continuity this will even hold, if one of A_1, B_1 is in M_1 and the other merely in $\mathcal{Q}(M_1)$ (and one of A_2, B_2 in M_2 , and the other merely in $\mathcal{Q}(M_2)$). So $\mathfrak{F}$ is a spatial isomorphism of $\mathcal{Q}(M_1)$ and $\mathcal{Q}(M_2)$ which carries $M_1^0, M_{1,0}$ into $M_2^0, M_{2,0}$ (by Corollary 1 to Theorem XI). Now (by the same corollary) $\mathfrak{F}_{M_2} \mathfrak{F} \mathfrak{F}_{M_1}^{-1}$ is a spatial isomorphism of $\mathfrak{H}_1$ and $\mathfrak{H}_2$ which carries M_1, M'_1 into M_2, M'_2 .

Theorem XI could be extended to cover cases (II_1, II_1) , where $C \gtrless 1$ in the standard normalization as well as other combinations of II_1 and II_∞ . In all cases a reduction of the spatial-isomorphism questions of $\mathfrak{H}_1$ and $\mathfrak{H}_2$ to algebraic-isomorphism questions of $\mathfrak{H}_1$ and $\mathfrak{H}_2$ (plus the behavior of C) results. We will discuss these questions in detail in later publications.

APPENDIX

1. Consider a ring M of class II_1 in $\mathfrak{H}$. Then M' is of class II_1 or II_∞ . Normalize D_M and $D_{M'}$, so as to have $D_M(\mathfrak{H}) = 1$ and $C = 1$, so $D_{M'}(\mathfrak{H}) = \alpha$ (cf. §1.1). By choosing an $n = 1, 2, \dots$ with $1/n \leq \alpha$ and then an $\mathfrak{W}' \eta M'$ with $D_{M'}(\mathfrak{W}') = 1/n$, apply R.O., §11.3, to form $M_{(\mathfrak{W}')} , M'_{(\mathfrak{W}')}$ in $\mathfrak{W}'$. Then $M_{(\mathfrak{W}')}$ is algebraically-ring-isomorphic to M , and $D_{M_{(\mathfrak{W}')}}(\mathfrak{W}') = D_M(\mathfrak{H}) = 1$ (cf. R.O., Lemmas 11.3.3 and 11.3.6). So if we are interested in the algebraical properties of M only, we may assume without any loss in generality that M, M' are in case II_1, II_1 and that $\alpha = 1/n$, $n = 1, 2, \dots$. Or, in standard normalization $C = n$, $n = 1, 2, \dots$ (cf. R.O., Theorem X).

Now form the direct product of $\mathfrak{H}$ with an n -dimensional Euclidean space $E_n \oplus \mathfrak{H}$ as in §2.1, for the case $C > 1$, and consider $M^{(2)}$ in $E_n \oplus \mathfrak{H}$. The argument used in §2.1 shows that $M^{(2)'} = R(N^{(1)}, M^{(2)'})$ and $C^{(2)} = C/n = 1$ and $M^{(2)}$ is ring isomorphic to M .

So we have for $M^{(2)}, M^{(2)'}$ in the standard normalization $C = 1$. If we are therefore interested in the algebraical properties of M only, we may even assume without any loss in generality that $C = 1$ in standard normalization; that is, $\alpha = 1$ in the normalization of §1.1. We will assume this in what follows.

It may be noticed concerning subrings that the metric of $\mathcal{Q}(M)$ (cf. Definition

4.3.1) is determined by the algebra of M (cf. §2.2). A subring demands closure in the weak operator topology, but it will be shown elsewhere that for subrings of M weak, relative closure in M for the $\|X - Y\|$ metric is equivalent to closure in the weak topology.

2. Under these conditions there is an analogy between M and the matrices of a Euclidean space E_n , described by an interesting Lebesgue-Stieltjes-Radon measure in the plane.

We can use the results of §§4.2–4.4, and thus we may form the Hilbert-space $Q(M)$ which is isomorphic to $\mathfrak{H}$, and in which M, M' are located by Theorem X.

Let $E(\lambda)$, $a < \lambda < b$, be a resolution of unity, all $E(\lambda) \in M$. Put $A = \int_{-\infty}^{\infty} \lambda dE(\lambda)$ (in the well known symbolic sense). For any Borel-set of real numbers S form

$$e_S(\lambda) = \begin{cases} 1 & \text{for } \lambda \in S, \\ 0 & \text{for } \lambda \notin S, \end{cases}$$

and

$$E(S) = e_S(A) = \int_{-\infty}^{\infty} e_S(\lambda) dE(\lambda) = \int_S dE(\lambda)$$

(symbolically). As $\overline{e_S(\lambda)} = e_S(\lambda) = (e_S(\lambda))^2$, so $e_S(A) = e_S(A)^* = (e_S(A))^2$; that is, $E(S) = e_S(A) \in M$ is a projection. (Cf. also Maeda, *Journal of Science, Hiroshima University*, Ser. A, vol. 4 (1934), pp. 57–91.)

Consider now point sets in the λ, μ -plane P and in particular sets of the form $S_1 \otimes S_2$:

$$(\lambda, \mu) \in S_1 \otimes S_2 \quad \text{means} \quad \lambda \in S_1, \mu \in S_2,$$

S_1, S_2 being two Borel-sets of real numbers. Define for any $X \in M$ (or even $X \in Q(M)$),

$$X_{S_1 \times S_2} = E(S_1)XE(S_2)$$

and

$$\begin{aligned} \omega(X; S_1 \otimes S_2) &= \|X_{S_1 \times S_2}\|^2 = \text{Tr}((X_{S_1 \times S_2})^* X_{S_1 \times S_2}) \\ &= \text{Tr}(E(S_2)X^*E(S_1) \cdot E(S_1)XE(S_2)) \\ &= \text{Tr}(E(S_2)X^* \cdot E(S_1)XE(S_2)) \\ &= \text{Tr}(E(S_1)XE(S_2) \cdot E(S_2)X^*) \\ &= \text{Tr}(E(S_1)XE(S_2)X^*). \end{aligned}$$

The first expression for $\omega(X, S_1 \otimes S_2)$ shows, that it is always ≥ 0 , the last one, that it is a totally additive set-function of S_1 and of S_2 .

Denote the set of all λ by Λ , then the above facts imply:

$$\begin{aligned}
 \omega(X; S_1 \otimes S_2) &\leq \omega(X; S_1 \otimes S_2) + \omega(X; (\Lambda - S_1) \otimes S_2) \\
 &= \omega(X; \Lambda \otimes S_2) \\
 &\leq \omega(X; \Lambda \otimes S_2) + \omega(X; \Lambda \otimes (\Lambda - S_2)) \\
 &= \omega(X; \Lambda \otimes \Lambda) = \llbracket X_{\Lambda \times \Lambda} \rrbracket^2 \\
 &= \llbracket 1 \cdot X \cdot 1 \rrbracket^2 = \llbracket X \rrbracket^2.
 \end{aligned}$$

So we have

Lemma A. (i) $\omega(X; S_1 \otimes S_2)$ is defined for all linear Borel-sets S_1, S_2 . (ii) It is totally additive in S_1 as well as in S_2 . (iii) We have always

$$0 \leq \omega(X; S_1 \otimes S_2) \leq \llbracket X \rrbracket^2 = \text{Tr}(X^* X).$$

3. We can use [Lemma A](#) to define a Lebesgue-Stieltjes-Radon measure $\mu(X; T)$ for all plane Borel-sets $T(\subset P)$ with the help of $\omega(X; S_1 \otimes S_2)$.

Definition A. If T is a plane Borel-set ($T \subset P$), then consider all sequences of linear Borel-set $S_1^{(i)}, S_2^{(i)}, i = 1, 2, \dots$ (all $S_1^{(i)}, S_2^{(i)} \subset \Lambda$) for which

$$(*) \quad T \subset \sum_{i=1}^{\infty} S_1^{(i)} \otimes S_2^{(i)}.$$

For every such sequence form the (numerical) sum

$$(**) \quad \sum_{i=1}^{\infty} \omega(X; S_1^{(i)} \otimes S_2^{(i)}).$$

Denote the g.l.b. of all numbers [\(**\)](#) by $\mu(X; T)$.

We prove now

Lemma B. (i) $\mu(X; T)$ is defined for all plane Borel-sets $T \subset P$.

(ii) It is totally additive in T .

(iii) We have always

$$0 \leq \mu(X; T) \leq \llbracket X \rrbracket^2 = \text{Tr}(X^* X).$$

(iv) In particular

$$\mu(X; S_1 \otimes S_2) = \omega(X; S_1 \otimes S_2)$$

and especially

$$\mu(X, P) = \llbracket X \rrbracket^2 = \text{Tr}(X^* X).$$

(i) is obvious. To prove (ii), observe first that our [Definition A](#) makes

$$\mu(X; T_1 + T_2 + \cdots) \leq \mu(X; T_1) + \mu(X; T_2) + \cdots$$

obvious, so we need to prove

$$\mu(X; T_1 + T_2 + \cdots) \geq \mu(X; T_1) + \mu(X; T_2) + \cdots$$

only when $T_i \cdot T_j = 0$ for $i \neq j$.

Even $\mu(X; T_1 + T_2) \geq \mu(X; T_1) + \mu(X; T_2)$ suffices. Then finite induction gives $\mu(X; T_1 + T_2 + \cdots + T_n) \geq \mu(X; T_1) + \mu(X; T_2) + \cdots + \mu(X; T_n)$, and so $\mu(X; T_1 + T_2 + \cdots) \geq \mu(X; T_1 + \cdots + T_n) \geq \mu(X; T_1) + \cdots + \mu(X; T_n)$, and as this holds for all $n = 1, 2, \dots$, it implies $\mu(X; T_1 + T_2 + \cdots) \geq \mu(X; T_1) + \mu(X; T_2) + \cdots$.

We will prove

$$(\S) \quad \mu(X; T) = \mu(X; TU) + \mu(X; T - TU)$$

for all T, U this gives our above inequality if $T = T_1 + T_2, U = T_1$. Call a U , for which [\(§\)](#) holds for all plane Borel-sets T , following Carathéodory (cf. [\(4\)](#), pp. 246–252), measurable. We must show that all U are measurable.

If $U = S_1 \otimes S_2$, then $T \subset \sum_{i=1}^{\infty} S_1^{(i)} \otimes S_2^{(i)}$ implies $TU \subset \sum_{i=1}^{\infty} S_1^{(i)} S_1 \otimes S_2^{(i)} S_2$, $T - TU \subset \sum_{i=1}^{\infty} S_1^{(i)} S_1 \otimes (S_2^{(i)} - S_2^{(i)} S_2) + \sum_{i=1}^{\infty} (S_1^{(i)} - S_1^{(i)} S_1) \otimes S_2^{(i)}$. Thus our [Definition A](#) gives immediately [\(§\)](#) with $\geq$ in it, and as $\leq$ is obvious (cf. above) it proves [\(§\)](#). So all $U = S_1 \otimes S_2$ are measurable, and therefore in particular all plane intervals (= rectangles).

Now the measurable sets U form a Borel-ring (cf., for instance, [\(4\)](#), loc. cit.). These considerations apply literally to the present case. Thus every plane Borel-set U is measurable. This completes the proof.

We now prove (iii). $\mu(X; T) \geq 0$ because all expressions [\(**\)](#) in [Definition A](#) are ≥ 0 . The relation $\mu(X; T) \leq \llbracket X \rrbracket^2 = \text{Tr}(X^* X)$ results by putting $S_1^{(1)} = S_2^{(1)} = \Lambda$ and all other $S_1^{(i)} = S_2^{(i)} = 0$.

To prove (iv), put $S_1^{(1)} = S_1, S_2^{(1)} = S_2$ and all other $S_1^{(i)} = S_2^{(i)} = 0$. This gives $\mu(X; S_1 \otimes S_2) \leq \omega(X; S_1 \otimes S_2)$. So we must prove $\geq$ only; that is,

$$S_1 \otimes S_2 \subset \sum_{i=1}^{\infty} S_1^{(i)} \otimes S_2^{(i)} \quad \text{implies} \quad \omega(X; S_1 \otimes S_2) \leq \sum_{i=1}^{\infty} \omega(X; S_1^{(i)} \otimes S_2^{(i)}).$$

Considering the properties of $\omega(X; S_1 \otimes S_2)$ given in [Lemma A](#), this implication follows literally as in the paper of [Łomnicki and Ulam](#) (*Fundamenta Mathematicae*, vol. 23 (1934), pp. 237–278; cf. also the lecture notes of the second-named author for the year 1934–1935).

So we have proved $\mu(X; S_1 \otimes S_2) = \omega(X; S_1 \otimes S_2)$. Put now $S_1 = S_2 = \Lambda$; then

$$\mu(X; P) = \mu(X; \Lambda \otimes \Lambda) = \omega(X; \Lambda \otimes \Lambda) = \llbracket X \rrbracket^2 = \text{Tr}(X^* X).$$

results.

4. The plane measure $\mu(X; T)$ may be used to “locate” the “position” of X in the λ, μ -plane P . It gives the entire plane a total “measure” or “weight” $\llbracket X \rrbracket^2$ (which is > 0 if $X \neq 0$), and any part T of it correspondingly a $\mu(X; T) (\geq 0)$. It plays the same role, as the sum of the absolute-value-squares of all matrix-elements in a certain area T in the matrix-scheme (which may be looked at as a region in the plane) for the matrices of a finite-dimensional Euclidean space E_n (that is, M in a case (I_n) , $n = 1, 2, \dots$).

We will analyse it more thoroughly in subsequent publications.

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